Hamiltonian and Hamilton-connected thresholds for planar recolouring graphs

Abstract

For a graph G, let $\mathcal{R}_k(G)$ have the proper colourings of G with a labelled k-colour palette as vertices, with one-vertex recolourings as edges. We prove that $\mathcal{R}_k(G)$ is Hamilton-connected and vertex-pancyclic whenever G is nonempty and d-degenerate and $k \geq d+3$; this uniform palette bound is sharp for both properties. We determine the common Hamiltonian and Hamilton-connected thresholds as 7 for planar graphs, 5 for triangle-free and bipartite planar graphs, and 4 for planar graphs of girth at least six and for forests. At four colours we also prove Hamiltonicity for planar graphs of girth at least five with at most fifteen distinct 5-cycles, for those whose blocks all have maximum degree at most three, and for those with at most one vertex of degree at least four. We also obtain Hamiltonicity for two-degenerate planar graphs of girth at least five with at most three vertices of degree at least four or at most seventeen distinct 5-cycles. The block condition allows unbounded degree at cutvertices. These four-colour proofs combine structural reductions with local restoration certificates supplied in a separate, reproducible supplement. For unrestricted planar graphs of girth at least five, both thresholds remain between 4 and 5.

1 Introduction

A cyclic listing of all proper colourings in which successive colourings differ at one vertex is a colouring Gray code, or equivalently a Hamilton cycle in the recolouring graph. Prescribing the first and last colourings asks for Hamilton-connectedness. Requiring a cycle of every length through each colouring asks for vertex-pancyclicity. We study these stronger spanning and cyclic properties for sparse graphs, with particular emphasis on sharp palette thresholds for planar classes.

All graphs are finite, simple and undirected, and may be disconnected. A graph is nonempty when it has at least one vertex. Put $[k] = \{1, \ldots, k\}$. The vertices of $\mathcal{R}_k(G)$ are the proper maps $\sigma : V(G) \to [k]$; two are adjacent when they differ on exactly one vertex of G. Colours are labelled, and a colouring need not use every colour. A proper colouring is called *frozen* when every vertex sees all the colours other than its own on its neighbourhood; such a colouring is an isolated vertex of the recolouring graph. A cycle has at least three vertices. A graph is Hamilton-connected if every two distinct vertices are joined by a spanning path, and vertex-pancyclic if every vertex lies on a cycle of every length from three to the order of the graph.

For a class $\mathcal{C}$ of nonempty graphs, define

$$h(\mathcal{C}) = \min\{q : \mathcal{R}_k(G) \text{ is Hamiltonian for every } G \in \mathcal{C} \text{ and every } k \geq q\},$$
$$\mathrm{hc}(\mathcal{C}) = \min\{q : \mathcal{R}_k(G) \text{ is Hamilton-connected for every } G \in \mathcal{C} \text{ and every } k \geq q\},$$

when these sets are nonempty. These are uniform thresholds in the palette size; the proofs establish every larger palette separately. A graph is d-degenerate if each nonempty subgraph has a vertex of degree at most d. Write $\mathcal{D}_d$ for this class.

Choo and MacGillivray [3, Theorem 3.4] proved that $\mathcal{R}_k(G)$ is Hamiltonian for $k \geq d+3$ when G is d-degenerate. They also determined the uniform forest threshold and several individual tree and cycle cases [3, Sections 5–6]. Our first result strengthens their general Hamiltonian conclusion while keeping the same palette bound.

Theorem 1.1. *Let G be a nonempty d-degenerate graph. For every $k \geq d+3$, $\mathcal{R}_k(G)$ is Hamilton-connected and vertex-pancyclic. Moreover, $\mathrm{hc}(\mathcal{D}_d) = d+3$, and $d+3$ is also the least uniform palette bound for vertex-pancyclicity on $\mathcal{D}_d$.*

Write $\mathcal{P}$ for the planar graphs, $\mathcal{T}$ for the triangle-free planar graphs, $\mathcal{B}$ for the bipartite planar graphs, $\mathcal{P}_g$ for the planar graphs of girth at least g, and $\mathcal{F}$ for the forests. All these classes contain only nonempty graphs; forests have infinite girth.

Theorem 1.2. *The common thresholds $h = \mathrm{hc}$ are 7 for $\mathcal{P}$, 5 for $\mathcal{T}$ and $\mathcal{B}$, and 4 for $\mathcal{P}_6$ and $\mathcal{F}$.*

The large-palette cases follow from degeneracy. The critical palettes require endpoint-preserving restoration of separated low-degree vertices, and, in the triangle-free case, of an adjacent degree-three pair. The lower bounds use frozen colourings and the three-colour recolouring graph of a short path. The equality $h(\mathcal{F}) = 4$ already follows from [3]; here it accompanies the Hamilton-connected forest result.

At four colours the girth-five case requires different local configurations. Let $\mathcal{P}_5^{\leq 15}$ consist of the graphs in $\mathcal{P}_5$ with at most fifteen distinct 5-cycles. Let $\mathcal{S}_5$ consist of those in $\mathcal{P}_5$ whose every block has maximum degree at most three. Here blocks include bridges and isolated vertices as well as maximal two-connected subgraphs.

Theorem 1.3. *For every $G \in \mathcal{P}_5^{\leq 15} \cup \mathcal{S}_5$ and every $k \geq 4$, $\mathcal{R}_k(G)$ is Hamiltonian. The same holds for $G \in \mathcal{P}_5$ with at most one vertex of degree at least four, and for two-degenerate $G \in \mathcal{P}_5$ with at most three such vertices or at most seventeen distinct 5-cycles. Each stated class has uniform Hamiltonian threshold four and Hamilton-connected threshold between four and five.*

In particular, Theorem 1.3 includes all subcubic planar graphs of girth at least five, while allowing vertices of arbitrarily large total degree when that degree is distributed among blocks. Bartier et al. [1] proved that $\mathcal{R}_4(G)$ is connected for every $G \in \mathcal{P}_5$. Connectivity supplies paths between states but does not supply a spanning cycle or prescribed endpoints, so these questions remain strictly stronger. For the full class we obtain

$$4 \leq h(\mathcal{P}_5) \leq 5, \qquad 4 \leq \mathrm{hc}(\mathcal{P}_5) \leq 5.$$

Table 1.1 records the results and these open cases.

1.1 Proof strategy and organization

Deleting vertices partitions the larger recolouring graph into fibres above colourings of the smaller graph. At the degenerate bound these fibres are cliques of order at least three. We choose bridge colours between consecutive fibres to prescribe endpoints or cycle lengths. Section 2 proves Theorem 1.1, including the parity obstruction that makes its bound sharp. Section 3 develops the product and independent-set lifts used in the planar proofs.

Section 4 proves Theorem 1.2 by combining these lifts with planar degree counts. Section 5 proves Theorem 1.3. The few-cycle argument combines the one-exceptional-vertex theorem with restoration of low-degree configurations, including paths whose endpoints share an outside neighbour and cycles with an attached degree-two vertex. A degree-two vertex also supplies a reservoir for an ordinary pentagon whose local fibre can be disconnected. For the subcubic, block-subcubic and one-exceptional-vertex classes, we restore two separated pentagons. Their product fibres have at most four components. Local cycles retain protected edges at each interface; a bounded-degree spanning tree then joins all component cycles into one spanning cycle, for a bottom recolouring cycle of arbitrary length.

Table 1.1: Uniform thresholds. An interval records an unresolved value.

Graph class	h	hc
Planar	7	7
Triangle-free planar	5	5
Bipartite planar	5	5
Planar, girth at least six	4	4
Forests	4	4
Planar, girth at least five, at most fifteen 5-cycles	4	$[4, 5]$
Planar, girth at least five, every block subcubic	4	$[4, 5]$
Planar, girth at least five, at most one vertex of degree at least four	4	$[4, 5]$
Two-degenerate planar, girth at least five, at most three vertices of degree at least four	4	$[4, 5]$
Two-degenerate planar, girth at least five, at most seventeen 5-cycles	4	$[4, 5]$
Planar, girth at least five	$[4, 5]$	$[4, 5]$

The supplement *Finite certificates for Hamiltonian thresholds in planar recolouring graphs* contains the literal paths, cycles, pools and interface records. Its Appendices A–G contain the finite data for these local restoration steps. Section 7 specifies exactly what the accompanying verifiers check. The certificate contract is deliberately finite: the readers certify (i) legality, coverage and adjacency of each listed local path or cycle, (ii) the stated symmetry orbits, (iii) the listed common interface edges, and (iv) the pool and protected-edge conditions in the finite component types. They do not certify the existence of a bottom Hamilton cycle, an arbitrary-length bad-run lift, a product cycle for an unlisted order, or a spanning-tree splice. Those steps are analytic arguments in the article, and the distinction is part of the claimed evidence rather than a hidden computational premise.

1.2 Notation

For a graph G, write $N_G(v)$ for the open neighbourhood, $N_G[v] = N_G(v) \cup \{v\}$, and $G - S$ for deletion of a vertex set S. The Cartesian product $X \square Y$ joins (x, y) to (x', y') when exactly one coordinate changes along an edge of its factor. A spanning path may be a singleton when its ends coincide in an intermediate construction. Local certificates use zero-based palettes $\{0, \dots, k-1\}$, obtained by relabelling $[k]$. Face lengths count edge-side incidences, so a bridge is counted twice.

2 Hamilton-connectedness and vertex-pancyclicity

We begin with the general result that supplies all large-palette statements.

Lemma 2.1. *Let $m \geq 3$ and let $L_1, \dots, L_m$ be finite sets of size at least two. Unless m is odd and all L_i are the same two-element set, there are $x_i \in L_i$ with $x_i \neq x_{i+1}$ cyclically.*

Proof. If some L_i has at least three elements, choose successively along the path obtained by deleting that position, then choose the remaining element to avoid both end values. If all sets are binary and two consecutive sets differ, start with a value in their symmetric difference and continue greedily. If all are the same binary set, alternation is possible exactly when m is even. □

2.1 Clique fibres

For a vertex v with $H = G - v \neq \varnothing$ and $k \geq d_G(v) + 3$, put

$$L(f) = [k] \setminus f(N_G(v)), \qquad F_f = \{f(c) : c \in L(f)\},$$

where $f(c)$ extends a proper colouring f of H by assigning colour c to v. The sets F_f partition $V(\mathcal{R}_k(G))$ into cliques, and adjacent base colourings f, g satisfy

$$|L(f)| \geq 3, \qquad |L(f) \cap L(g)| \geq |L(f)| - 1 \geq 2. \tag{2.1}$$

The cross edges are exactly $f(c)g(c)$ for $c \in L(f) \cap L(g)$. If the base move changes a nonneighbour of v, then $L(f) = L(g)$. We use these fibres throughout the single-vertex lifts.

Lemma 2.2. *Let $H = G - v \neq \varnothing$ and $k \geq d_G(v) + 3$. If $f_0, f_1, \ldots, f_{s-1}, f_0$ is a cycle of $\mathcal{R}_k(H)$ and $x \in F_{f_0}$, then $\mathcal{R}_k(G)$ has a cycle through x of every length in*

$$[2s + 1,\ \textstyle\sum_i |F_{f_i}|]. \tag{2.2}$$

In particular, if $\mathcal{R}_k(H)$ is Hamiltonian, then so is $\mathcal{R}_k(G)$.

Proof. Put $S_i = L(f_i) \cap L(f_{i+1})$, with cyclic indices. Lemma 2.1 selects bridge colours $y_i \in S_i$ with $y_{i-1} \neq y_i$, unless s is odd and all S_i equal one binary set $\{p, q\}$.

We exclude this exception. By (2.1), each $L(f_i)$ then has size three, and

$$k - 3 = |f_i(N_G(v))| \leq d_G(v) \leq k - 3.$$

Thus the $k-3$ neighbours of v have distinct colours, none uses p or q, and every base move changes a neighbour of v; a nonneighbour move would leave an intersection of size three. Order the neighbours $u_1, \ldots, u_{k-3}$. In the $(k-2)$-element palette $[k] \setminus \{p, q\}$, append the unique unused colour r_i to the tuple $(f_i(u_1), \ldots, f_i(u_{k-3}))$. A base move exchanges one neighbour colour with r_i, reversing permutation parity. An odd number of such exchanges cannot return to the initial tuple. Hence the bridge colours exist.

Choose an m_0-vertex path in F_{f_0} between its bridge states and containing x, with $3 \leq m_0 \leq |F_{f_0}|$. In every other clique choose a bridge-to-bridge path on $2 \leq m_i \leq |F_{f_i}|$ vertices. The sums of these integer intervals are exactly (2.2). The cross edges with v fixed at y_i close these paths into the desired cycle. A Hamilton base cycle with every $m_i = |F_{f_i}|$ gives a Hamilton cycle. □

Lemma 2.3. *Let $t \geq 1$, and let $S_0, \ldots, S_{t-1}$ be finite sets of colours such that $|S_i| \geq 2$ for every i. Suppose that $|S_j| \geq 3$ for some $j \in \{0, \ldots, t-1\}$. For any colours a and b, there exist colours $x_i \in S_i$ satisfying*

$$x_0 \neq a, \qquad x_i \neq x_{i-1} \quad (1 \leq i \leq t-1), \qquad x_{t-1} \neq b.$$

Proof. Set $x_{-1} = a$ and $x_t = b$. Choose $x_0, \ldots, x_{j-1}$ forwards, avoiding the preceding colour, and $x_{t-1}, \ldots, x_{j+1}$ backwards, avoiding the following colour. Each choice excludes only one colour. Finally choose $x_j \in S_j \setminus \{x_{j-1}, x_{j+1}\}$, which is possible since $|S_j| \geq 3$. The boundary assignments also cover $j = 0$, $j = t-1$, and $t = 1$. □

Lemma 2.4. *Let $H = G - v \neq \varnothing$ and $k \geq d_G(v) + 3$. Suppose $Y = \mathcal{R}_k(H)$ is nonempty and Hamilton-connected, and v has a nonneighbour in H. If for every $f \in V(Y)$ and $c \in L(f)$ there is a neighbour g of f with $c \in L(g)$, then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. Fix a nonneighbour w of v. Colour permutations give all k colours at w among the states of Y; deleting at most one state leaves at least $k-1 \geq 2$ colours. Thus every base path missing at most one state contains a move at w. Its consecutive intersections $S_i = L(f_i) \cap L(f_{i+1})$ have size at least two, and the move at w supplies one of size at least three. Lemma 2.3 now gives bridge colours avoiding any prescribed first and last colours, with distinct entrance and exit colours in every clique. Hamilton paths inside the cliques concatenate to a spanning path through those fibres with the prescribed endpoints.

Now prescribe distinct endpoints $\alpha(a)$ and $\beta(b)$ in $\mathcal{R}_k(G)$. If $\alpha \neq \beta$, lift a Hamilton path in Y between their restrictions.

If $\alpha = \beta$, choose $c \in L(\alpha) \setminus \{a, b\}$ and a neighbour g preserving c. Since $|V(Y)| \geq k \geq 3$, closing a Hamilton path between the ends of αg gives a Hamilton cycle through that edge. Let h be its other neighbour of α and choose $q \in (L(\alpha) \cap L(h)) \setminus \{c\}$. Since q cannot equal both a and b, and a spanning path may be reversed, we may interchange a and b so that $q \neq a$.

Partition F_α into paths $P : \alpha(a) \rightsquigarrow \alpha(c)$ and $P' : \alpha(q) \rightsquigarrow \alpha(b)$. Use a singleton for P' if $q = b$, and its endpoint edge otherwise; put all remaining vertices in P. Deleting α from the chosen base cycle leaves a path missing only that state. Lift it from $g(c)$ to $h(q)$ through all other fibres. The cross edges at c, q join this lifted path to P, P', giving the prescribed spanning path. □

Theorem 2.5. *Let G be a nonempty d-degenerate graph and $k \geq d+3$. Then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. Induct on $n = |V(G)|$, starting with $\mathcal{R}_k(K_1) = K_k$. If $G = K_n$, then $\mathcal{R}_k(G)$ is the arrangement graph $A_{k,n}$. Here $A_{k,n}$ is the graph of ordered n-tuples of distinct elements of $[k]$, with one-coordinate changes; this is the parameter convention used in [5]. Since $k \geq n+2$, it is Hamilton-connected by [5, Theorem 1].

Otherwise a minimum-degree vertex v has degree at most d and a nonneighbour. Put $H = G - v$; induction makes $\mathcal{R}_k(H)$ Hamilton-connected. For each base colouring f and $c \in L(f)$, choose a vertex of degree at most d in H. It has at least three available colours, so can be recoloured avoiding its present colour and c. This base move preserves c at v. Lemma 2.4 applies. □

Theorem 2.6. *Let G be a nonempty d-degenerate graph and $k \geq d+3$. Then $\mathcal{R}_k(G)$ is vertex-pancyclic.*

Proof. Induct on $|V(G)|$, starting with $\mathcal{R}_k(K_1) = K_k$. For larger G, choose v with $d_G(v) \leq d$, put $H = G - v$, use the clique fibres above, and write $Y = \mathcal{R}_k(H)$, $M = |V(Y)|$, and $N = |V(\mathcal{R}_k(G))|$. Colour permutations give $M \geq k \geq 3$. Fix a state $x = \alpha(a)$ of $\mathcal{R}_k(G)$.

The clique F_α already supplies a triangle through x. For lengths four, five and six, recolour a vertex of degree at most d in H to a colour avoiding its present colour, its neighbours, and a. At most $d+2$ colours are excluded, so this gives a neighbour β of α with $a \in L(\beta)$. Choose $p \in (L(\alpha) \cap L(\beta)) \setminus \{a\}$, $q \in L(\alpha) \setminus \{a, p\}$, and $r \in L(\beta) \setminus \{a, p\}$. The following cyclic sequences have the required lengths:

$$\begin{aligned} 4: &\quad \alpha(a), \alpha(p), \beta(p), \beta(a), \\ 5: &\quad \alpha(a), \alpha(q), \alpha(p), \beta(p), \beta(a), \\ 6: &\quad \alpha(a), \alpha(q), \alpha(p), \beta(p), \beta(r), \beta(a). \end{aligned}$$

For each $3 \leq s \leq M$, induction supplies a base s-cycle C_s through α. Put $T_s = \sum_{f \in V(C_s)} |F_f|$. Lemma 2.2 gives every length in $[2s+1, T_s]$. These intervals overlap successively, since $T_s \geq 3s \geq 2s+3$ for $s \geq 3$. Also $T_M = N$. They therefore cover every length from seven to N, completing the proof. □

Remark 2.7. Vertex-pancyclicity here cannot be strengthened to edge-pancyclicity (cycles of every length through each edge). For any $k \geq 4$, colour the leaves of the 1-degenerate star $G = K_{1,k-2}$ distinctly with $3, \ldots, k$. The extensions with centre colours 1 and 2 form an edge of $\mathcal{R}_k(G)$ in no triangle: such a triangle would recolour only the centre, which has no third available colour.

Proposition 2.8. *For every $d \geq 0$, $\mathrm{hc}(\mathcal{D}_d) = d + 3$. The same uniform palette bound for vertex-pancyclicity cannot be reduced.*

Proof. The upper bounds are the preceding theorems. For $d \geq 1$, take $G = K_{d+1}$ and $k = d + 2$. Append the unique unused colour to each ordered colouring of G. The resulting permutation of $d + 2$ colours changes by a transposition at every recolouring. Thus $\mathcal{R}_{d+2}(G)$ is bipartite by permutation parity, with equal parts and even order $(d + 2)! \geq 6$. A spanning path must have endpoints in opposite parts, so the graph is not Hamilton-connected. It has no triangle, so it is not vertex-pancyclic. For $d = 0$, the two-isolated-vertex graph has $\mathcal{R}_2(G) = C_4$, which fails both properties. □

3 Endpoint lifting and restoration

A colouring of a deleted graph determines the available colours at the vertices to be restored. The resulting extension graph is a fibre. In all the constructions below, assignments retain their actual coordinate values: equal assignments in two different fibres are distinct vertices, and a cross edge joins their two copies. In particular, a move of one vertex of the deleted graph may change several available lists at once.

Lemma 3.1. *Let $s \geq 1$ and $F = \square_{j=1}^{s} K_{A_j}$, where every A_j is finite and $|A_j| \geq 2$. If all lists have size two, F has a Hamilton path between two prescribed vertices if and only if those vertices lie in opposite bipartition classes. If at least one list has size at least three, F is Hamilton-connected.*

Proof. In the binary case F is the hypercube Q_s, whose bipartition classes have equal size. The endpoint condition is therefore necessary. For sufficiency, induct on s, starting with the single edge Q_1. Choose a coordinate on which the prescribed endpoints differ and split along it. Their projections into Q_{s-1} have the same parity. Choose a projected vertex of the opposite parity; use the two induction paths to and from its copies, joined by the edge between those copies.

For the second assertion, we first show that $Y \square K_r$ is Hamilton-connected whenever Y is Hamilton-connected and $|V(Y)| \geq 3$. Induct on r, starting with $r = 1$. If the two endpoints lie in different copies of Y, order the copies starting and ending at those copies. Choose successive bridge vertices in $V(Y)$, with the first different from the first endpoint projection, consecutive bridges different, and the last different from the last endpoint projection. Such choices exist because $|V(Y)| \geq 3$: choose greedily, leaving the last bridge until the end, when it has at most two forbidden values. Prescribed-end Hamilton paths inside the copies then join into a spanning path. If both endpoints lie in one copy, take a Hamilton path between them there and choose an edge xy of it. By induction the other $r - 1$ copies have a spanning path from a copy of x to a copy of y in the same one of those copies. Insert that path in place of xy, using the two cross edges. Start with a complete-graph factor of order at least three and add the remaining factors to obtain the claim. □

Lemma 3.2. *Let $m \geq 3$, $s \geq 1$, and let*

$$F_i = \square_{j=1}^{s} K_{A_{i,j}}, \qquad |A_{i,j}| \geq 2, \qquad 0 \leq i < m,$$

be disjoint fibres arranged along a path. Join equal assignments in consecutive fibres. At every transition assume that at least one list is unchanged and that, for every coordinate,

$$|A_{i,j} \setminus A_{i+1,j}| \le 1, \qquad |A_{i+1,j} \setminus A_{i,j}| \le 1.$$

If an internal fibre F_r, $1 \le r \le m-2$, has a factor of order at least three, then every prescribed $p \in F_0$ and $q \in F_{m-1}$ are joined by a path covering all vertices of all fibres exactly once.

Proof. Write

$$J_i = \square_{j=1}^{s} K_{A_{i,j} \cap A_{i+1,j}}$$

for the graph of common assignments. The intersections are nonempty, and an unchanged coordinate supplies an edge of J_i.

Starting at p, record the reachable exit assignments at each interface. If a fibre has a reachable entrance a, an interface edge has an admissible Hamilton-path endpoint: in a binary fibre choose its endpoint of parity opposite to a, and in a nonbinary fibre choose an endpoint different from a. Lemma 3.1 therefore shows that the reachable set never becomes empty.

At F_r, its outgoing interface J_r still contains an edge after any one assignment is removed. Indeed, if its large coordinate is unchanged, J_r contains a triangle. Otherwise that coordinate has intersection size at least two, and a different unchanged coordinate supplies a $K_2 \square K_2$. Take a reachable entrance a in F_r. Hamilton-connectedness of F_r makes every assignment in $V(J_r) \setminus \{a\}$ reachable; this set contains an entire edge.

An incoming reachable edge permits every outgoing assignment to be reached in the next fibre. In a binary fibre the two endpoints of the edge have opposite parities, so one has the parity needed for any specified outgoing assignment. In a nonbinary fibre they are distinct, so one differs from that assignment. Thus the reachable exits contain an edge at every subsequent interface. At the last fibre, one endpoint of the incoming edge can be joined to the specified q. Backtracking the finite choices supplies one Hamilton path in each fibre; the actual cross edges concatenate them into the required spanning path. □

Lemma 3.3. *Let S be a nonempty independent set, $H = G - S \neq \varnothing$, and $k \ge 4$. Suppose*

$$d_G(u) \le k-2 \quad (u \in S), \qquad |N_G(x) \cap S| \le |S| - 1 \quad (x \in V(H)).$$

Assume that $\mathcal{R}_k(H)$ is Hamilton-connected and that some proper k-colouring of H leaves at least three colours available at a vertex of S. If α, β are proper k-colourings of G with $\alpha|_H \neq \beta|_H$, then $\mathcal{R}_k(G)$ has an α–β Hamilton path.

Proof. Take a Hamilton path of $\mathcal{R}_k(H)$ between the two restriction colourings. For a base colouring f, the fibre is exactly

$$\square_{u \in S} K_{A_f(u)}, \qquad A_f(u) = [k] \setminus f(N_G(u)).$$

Every list is at least binary. A move at an actual vertex x of H leaves the list of at least one non-neighbour in S unchanged; every other list loses and gains at most one colour. These are the transition conditions of Lemma 3.2, even when several lists change together.

A colouring with a nonbinary fibre has at least k distinct colour permutations: fix a vertex of the nonempty H and send its colour to each of the k labels. Every resulting fibre is nonbinary. The base Hamilton path has only two endpoints, so one of these colourings is internal. Lemma 3.2 now lifts the path with the specified, actual colours of α and β on S. □

For use of this lemma, first choose a vertex w on which two prescribed colourings differ, and then choose S with $w \notin S$. This ensures different restriction colourings; the lemma does not claim to handle arbitrary endpoints in the same fibre.

Lemma 3.4. *Let $m \geq 3$ disjoint fibres $F_i = \square_{j=1}^{s} K_{A_{i,j}}$, with $s \geq 1$ and $|A_{i,j}| \geq 2$, be cyclically ordered. Join equal assignments in consecutive fibres and define $J_i = \square_j K_{A_{i,j} \cap A_{i+1,j}}$, with indices modulo m. Suppose every J_i contains an edge, F_0 has a factor of order at least three, and there are distinct edges $e_0 \in E(J_0)$ and $e_{m-1} \in E(J_{m-1})$ (compared as actual assignment pairs). Then the layered graph has a Hamilton cycle.*

Proof. Choose an edge e_i in every remaining interface, and choose $z_0 \in V(e_0) \setminus V(e_{m-1})$. For $1 \leq i < m$, there is a bijection $\phi_i : V(e_{i-1}) \to V(e_i)$ such that each $z, \phi_i(z)$ is an admissible pair of Hamilton-path endpoints in F_i. In a binary fibre match opposite parities. Otherwise use a bijection with no fixed assignment, which always exists between two two-element sets: interchange equal sets, send a common member to the other target member when there is exactly one common member, and use either bijection when the sets are disjoint.

Put $z_i = \phi_i(z_{i-1})$. Traverse F_i from z_{i-1} to z_i for $1 \leq i < m$. Since $z_{m-1} \in V(e_{m-1})$, $z_{m-1} \neq z_0$, so the nonbinary F_0 has a Hamilton path from z_{m-1} to z_0. Join the fibre paths by the cross edges at the assignments z_i. They cover every fibre once and form one cycle. □

Lemma 3.5. *Let S be a nonempty independent set, $H = G - S \neq \varnothing$, and $k \geq 4$. Suppose*

$$d_G(u) \leq k - 2 \quad (u \in S), \qquad |N_G(x) \cap S| \leq |S| - 1 \quad (x \in V(H)).$$

If $\mathcal{R}_k(H)$ is Hamiltonian and some proper k-colouring of H leaves at least three colours available at a vertex of S, then $\mathcal{R}_k(G)$ is Hamiltonian.

Proof. Take a Hamilton cycle of the base and use the same product fibres as in Lemma 3.3. Every interface contains an edge. Index a nonbinary fibre as F_0 and choose a large coordinate in it. At either adjacent interface, that coordinate supplies a triangle if unchanged, or a $K_2 \square K_2$ together with a different unchanged coordinate if it changes. Thus both adjacent interfaces have at least two edges, and distinct edges e_0, e_{m-1} can be chosen. Apply Lemma 3.4. □

Lemma 3.6. *Let $H = G - v \neq \varnothing$ and $k \geq d_G(v) + 3$. If v has a nonneighbour, $\mathcal{R}_k(H)$ is Hamilton-connected, and $\delta(\mathcal{R}_k(H)) > d_G(v)$, then $\mathcal{R}_k(G)$ is Hamilton-connected.*

Proof. For each base colouring f and $c \in L(f)$, at most $d_G(v)$ moves make c unavailable at v: only a neighbour of v recoloured to c can do so. The state-degree bound therefore supplies a neighbour preserving c. Apply Lemma 2.4. □

A Hamilton-connected graph of order at least four has minimum degree at least three. A vertex of degree two would, with its neighbours chosen as the two endpoints, force a spanning path to have only three vertices; degrees zero and one are excluded as well. Every nonempty colourable H has at least k states in $\mathcal{R}_k(H)$, by colour permutation. These observations will supply the state-degree condition in the low-degree inductions.

4 Hamilton-connected planar classes

4.1 All planar graphs

We first establish the marked structural statement needed to retain a coordinate on which the prescribed endpoint colourings differ.

Lemma 4.1. *If G is planar and $\delta(G) \geq 5$, then, for any $w \in V(G)$, there are two degree-five vertices different from w whose distance is at least three. Vertices in different components have infinite distance.*

Proof. Extend G, without adding vertices, to a simple plane triangulation T. A degree-five vertex in T still has degree five in G, and a pair at distance at least three in T has that property in G. Moreover $\delta(T) \geq 5$. It suffices to prove the assertion in T. Euler's formula gives at least twelve degree-five vertices.

For a degree-five vertex x, put $W = N_T[x]$, $U = V(T) \setminus W$, $r = |U|$, and let B be the vertices at distance exactly two from x; write $b = |B|$. The five neighbours of x form a cycle in their cyclic order, so $e_T(W) \geq 10$. Contract the connected W and simplify parallel edges. The resulting plane graph has $r + 1$ vertices and $e_T(U) + b$ edges. As $|V(T)| \geq 12$, the planar edge bound gives $e_T(U) + b \leq 3r - 3$. Since $e(T) = 3r + 12$,

$$\sum_{z \in U} d_T(z) = e_T(U) + e(T) - e_T(W) \leq 6r - b - 1.$$

Let F_x be the degree-five vertices at distance at least three from x. The same sum is exactly

$$6r - b - |F_x| + \sum_{z \in B} (d_T(z) - 5) + \sum_{\substack{z \in U \setminus B \\ d_T(z) \geq 6}} (d_T(z) - 6).$$

It follows that $|F_x| \geq 1$, and if $|F_x| = 1$, every vertex of B has degree five.

Suppose every pair of degree-five vertices at distance at least three contains w, and let L be all degree-five vertices except w. For every $x \in L$, the nonempty F_x must be $\{w\}$. Hence w is at distance at least three from L, and every vertex at distance exactly two from x belongs to L. Take a shortest path from w to L, ending at x. The vertex two steps before x is also in L and is closer to w, a contradiction. □

Theorem 4.2. *For every nonempty planar graph G and every $k \geq 7$, $\mathcal{R}_k(G)$ is Hamilton-connected. Consequently $h(\mathcal{P}) = \mathrm{hc}(\mathcal{P}) = 7$.*

Proof. Prove the seven-colour assertion by induction on $|V(G)|$, allowing disconnected graphs throughout. If $|V(G)| \leq 5$, the graph is four-degenerate and Theorem 2.5 applies. For $|V(G)| \geq 6$, suppose first that some vertex v has degree at most four; it has a nonneighbour. The smaller $H = G - v$ is Hamilton-connected by induction. For every seven-colouring f of H, which has at least five vertices,

$$d_{\mathcal{R}_7(H)}(f) = \sum_{x \in V(H)} (6 - |f(N_H(x))|) \geq 6|V(H)| - 2|E(H)| \geq 12 > d_G(v).$$

Lemma 3.6 restores v.

Otherwise $\delta(G) \geq 5$. For prescribed distinct colourings α, β, choose an actual vertex w with $\alpha(w) \neq \beta(w)$. Lemma 4.1 supplies $S = \{u, v\}$, avoiding w, with $d(u) = d(v) = 5$ and $d_G(u, v) \geq 3$. Thus S is independent, no outside vertex is adjacent to both members, and $H = G - S$ is smaller and nonempty. Choose two nonadjacent neighbours x, y of u; they exist since its five neighbours cannot induce a planar K_5. Properly six-colour the planar $H - \{x, y\}$, and give both x, y colour seven. This is a proper colouring of H in which the neighbours of u use at most four colours, leaving at least three available. The restriction colourings differ at w, so Lemma 3.3 gives an α–β Hamilton path.

For $k \geq 8$, planar five-degeneracy and Theorem 2.5 suffice. For sharpness, use the following explicit model of the icosahedron. Put $\varphi = (1+\sqrt{5})/2$ and take the twelve points

$$\{(0, \pm 1, \pm\varphi), (\pm 1, \pm\varphi, 0), (\pm\varphi, 0, \pm 1)\},$$

joining two points at distance 2. The six antipodal pairs are independent, and a direct check of the five neighbours of any point shows that they contain exactly one point from each of the other five pairs. Colour each antipodal pair with its own colour. Every vertex then sees all other five colours, so the colouring is frozen: no one-vertex recolouring is possible. Relabelling the six colours gives distinct frozen states, so the recolouring graph is disconnected. This is the standard icosahedral obstruction (see also [2]), and proves that both thresholds are seven. □

Planar five-degeneracy and Theorem 2.6 also give vertex-pancyclicity for $k \geq 8$.

4.2 Triangle-free and bipartite planar graphs

The critical five-colour step needs an adjacent-pair fibre, which is not a product of complete graphs. We give its finite input separately from the arbitrary-size lifting argument.

For $A, B \subseteq \{0,1,2,3,4\}$, put

$$F(A,B) = \{(a,b) \in A \times B : a \neq b\},$$

with adjacency given by changing one coordinate. When $|A|, |B|$ are three or four, exchange the coordinates if necessary so that $|A| \leq |B|$. There are exactly seven types $(|A|, |B|, |A \cap B|)$.

Lemma 4.3. *For each of the seven types*

$$331, 332, 333, 342, 343, 443, 444,$$

every edge of $F(A,B)$ lies on a Hamilton cycle. The type 333 is itself a 6-cycle. The types $342, 343, 443, 444$ are Hamilton-connected. If J is the common state set of one of the interfaces allowed in Lemma 4.4, then every $p \in F(A,B)$ has a neighbour $q \in J$, and hence a p–q Hamilton path in $F(A,B)$.

Proof. Supplementary Appendix A gives one short Hamilton cycle for each of the seven representatives and checks its orbit under the independent row and column permutations that agree on $A \cap B$ (with the stated side exchange for type 332). The resulting edge orbits are the full edge sets. The 53 endpoint words in supplementary Appendix A prove the four Hamilton-connected types. Their symmetry coverage and an exact subset dynamic program verify the complete endpoint classification.

For the interface exit, exchange coordinates if needed so that B is unchanged, and put $A_0 = A \cap A'$. All allowed transitions, including the unchanged case, give $|A_0| \geq 2$ and $|B| \geq 3$. For $p = (a,b)$ with $a \in A_0$, choose (a,d) with $d \in B \setminus \{a,b\}$; otherwise choose (c,b) with $c \in A_0 \setminus \{b\}$. Either choice is a neighbour in J. Deleting this edge from a Hamilton cycle gives the required p–q Hamilton path. □

Lemma 4.4. *Arrange full fibres $F(A_i, B_i)$ along a base path. Each list has size three or four in a fixed five-colour palette. At each step either both lists stay unchanged, or exactly one changes. A changed list is either replaced by another three-element list differing in one colour, or changes between a three-element list and a containing four-element list. Join the copies of equal assignments at consecutive layers. If an internal layer has $|A_r| = |B_r| = 4$, there is a spanning path between any prescribed first-layer and last-layer states.*

Proof. First consider one allowed interface between $F(A, B)$ and $F(A', B')$, and write J for their common assignment set. The preceding lemma supplies, in both directions, a Hamilton path from every entrance state to some state of J.

Use this property to propagate nonempty sets of reachable exits from the prescribed first state. The internal 44 layer is Hamilton-connected, and its outgoing interface has at least nine states. Any reachable entrance p can therefore reach all interface states except possibly p, yielding at least two distinct reachable exits. The next layer still has one four-element list and is one of $342, 343, 443, 444$, hence is Hamilton-connected. Two distinct reachable entrances let every state of that layer be an exit. If it is the last layer, this includes the prescribed endpoint. Otherwise its entire outgoing interface is reachable.

Thereafter this property propagates: for a specified exit q in the current layer, apply the one-interface property in reverse to find a Hamilton-path entrance in its fully reachable incoming interface. The last prescribed endpoint is therefore reachable. Backtracking gives one Hamilton path in every full fibre, joined by actual cross edges, with no repetition or omission. □

Theorem 4.5. *If G is a nonempty triangle-free planar graph, then $\mathcal{R}_k(G)$ is Hamilton-connected for every $k \geq 5$. For the class of nonempty triangle-free planar graphs, $h = \mathrm{hc} = 5$.*

Proof. For $k \geq 6$, use three-degeneracy and Theorem 2.5. For $k = 5$, induct on the order of G, permitting disconnected graphs. Orders at most three are handled by two-degeneracy and the same theorem. At larger orders, suppose first that some vertex v has degree at most two; it has a nonneighbour. The induction hypothesis gives a Hamilton-connected base of at least five states, thus of minimum degree at least three; Lemma 3.6 applies.

Assume now $\delta(G) \geq 3$. Let L be its degree-three vertices and $s = |L|$. The triangle-free planar edge bound gives

$$s \geq 8 + \sum_{d_G(z) \geq 4} (d_G(z) - 4).$$

Given distinct α, β, choose w on which they differ. There are two exhaustive cases.

If $L \setminus \{w\}$ contains adjacent u, v, denote their other neighbours by u_1, u_2 and v_1, v_2. The four actual outside vertices are distinct: a common neighbour of u and v would form a triangle. In $H = G - \{u, v\}$ the restriction endpoints still differ at w, so induction gives a base Hamilton path between them. Its fibre over f is

$$F(A_f, B_f), \qquad A_f = [5] \setminus \{f(u_1), f(u_2)\}, \quad B_f = [5] \setminus \{f(v_1), f(v_2)\}.$$

The lists have size three or four. One actual base move affects at most one of them and gives precisely an interface allowed in Lemma 4.4. Edges among the four outside vertices restrict the base colourings, but do not alter this statement.

There is a base colouring with both lists four-element: use Grötzsch's theorem [4] to three-colour $H - \{u_1, u_2, v_1, v_2\}$, then give u_1, u_2 colour four and v_1, v_2 colour five. Each pair is independent by triangle-freeness, and edges across the pairs have distinct colours. At least five global colour permutations give distinct double-four-list base states. Since the Hamilton path has only two endpoints, one is internal. Apply Lemma 4.4 with the specified local endpoint states to complete the restoration.

Otherwise $S = L \setminus \{w\}$ is independent and has at least $s - 1 \geq 7$ vertices; put $H = G - S$. For any outside vertex z of degree at least four, the displayed Euler budget implies $d_G(z) \leq s - 4 < |S|$. The only possible degree-three outside vertex is w, whose degree is also less than $|S|$. Thus no outside vertex sees all of S. Choose $u \in S$ and two of its neighbours x, y. They are nonadjacent. Three-colour $H - \{x, y\}$ by Grötzsch's theorem and give both x, y colour five. At most two colours

then occur on the three neighbours of u, so u has at least three available colours. All conditions of Lemma 3.3 hold, and w is retained; it gives the desired endpoint path.

For sharpness, on the cube Q_3 view the colours as $\mathbb{Z}_2^2$ and set

$$f(x, y, z) = x(1, 0) + y(0, 1) + z(1, 1).$$

The three neighbours of every vertex have all three other colours, so the colouring is frozen and the complete four-colour recolouring graph is disconnected. The upper and lower bounds give both thresholds. □

Corollary 4.6. *The class of nonempty bipartite planar graphs has $h = \mathrm{hc} = 5$. For every graph in this class, $\mathcal{R}_k(G)$ is vertex-pancyclic for $k \geq 6$.*

Proof. Bipartite planar graphs are triangle-free, and the cube belongs to this class. The vertex-pancyclic assertion follows from three-degeneracy and Theorem 2.6. □

The same vertex-pancyclic bound $k \geq 6$ holds for all triangle-free planar graphs. This conclusion uses the degenerate theorem separately from Hamilton-connectedness.

4.3 Girth at least six and forests

Lemma 4.7. *If G is planar, $\mathrm{g}(G) \geq 6$, and $\delta(G) \geq 2$, then for any $w \in V(G)$ there are two degree-two vertices different from w whose distance is at least three.*

Proof. Every component has a cycle, and the girth-six Euler bound gives at least six degree-two vertices per component. If there are at least two components, choose such a vertex different from w in each of two components. Suppose G is connected and let n_2 be its number of degree-two vertices. From $2|E(G)| \leq 3|V(G)| - 6$,

$$n_2 \geq 6 + \sum_{d_G(z) \geq 3} (d_G(z) - 3).$$

Let L consist of all degree-two vertices except w; then $|L| \geq n_2 - 1 \geq 5$. Suppose all pairs in L have distance at most two.

If $u, v \in L$ are adjacent, let a, b be their respective other neighbours. The absence of cycles of length at most five implies that the intersection of the two closed distance-two balls is exactly $\{u, v, a, b\}$, and that $d_G(a, b) = 3$. Thus at most three of these vertices can lie in L, a contradiction.

Otherwise L is independent. Its two-element neighbourhoods are pairwise intersecting; they are distinct because there is no four-cycle. At least four distinct pairwise intersecting two-element sets have a common element: if $\{a, b\}, \{a, c\}$ occur and a third set omits a, it must be $\{b, c\}$, after which no fourth distinct set is possible. Hence some vertex z is adjacent to all of L. Then $d_G(z) \geq n_2 - 1 \geq 5$, whereas the Euler inequality gives

$$n_2 \geq 6 + d_G(z) - 3 \geq n_2 + 2,$$

a contradiction. □

Theorem 4.8. *For every finite integer $g \geq 6$, the class of nonempty planar graphs of girth at least g has $h = \mathrm{hc} = 4$. Every graph G in this class has Hamilton-connected $\mathcal{R}_k(G)$ for $k \geq 4$, and vertex-pancyclic $\mathcal{R}_k(G)$ for $k \geq 5$.*

Proof. For $k \geq 5$, use two-degeneracy and Theorems 2.5 and 2.6. At $k = 4$, induct on the order, starting with orders at most two, which are forests and covered by the one-degenerate theorem. If $|V(G)| \geq 3$ and $d_G(v) \leq 1$, then v has a non-neighbour. The smaller nonempty base is Hamilton-connected by induction and has at least four states, hence minimum degree at least three. Restore v using Lemma 3.6.

Otherwise apply Lemma 4.7 to a vertex w at which the prescribed endpoint colourings differ. It gives separated degree-two vertices u, v, both different from w. Set $S = \{u, v\}$ and $H = G - S$. The set S is independent, no outside vertex sees both its members, and H is nonempty. The neighbours x, y of u are nonadjacent. The two-degenerate $H - \{x, y\}$ can be three-coloured; giving x, y colour four leaves three colours available at u. Lemma 3.3 now restores a Hamilton path between the actual prescribed states.

For the lower bound, take a cycle of length at least g divisible by three and repeat the colours $0, 1, 2$ around it. This is a frozen proper three-colouring, so its complete three-colour recolouring graph is disconnected. Both class thresholds are therefore four. □

Theorem 4.9. *The class of nonempty forests has $h = \mathrm{hc} = 4$. Every such forest F has Hamilton-connected and vertex-pancyclic $\mathcal{R}_k(F)$ for $k \geq 4$.*

Proof. The upper bounds follow from one-degeneracy and Theorems 2.5 and 2.6. For the Hamiltonian lower bound, $\mathcal{R}_3(P_3)$ has twelve states. Its six states using three distinct colours have degree two, so every edge incident with them is forced in a spanning cycle. For each of the three middle-vertex colours b, these forced edges form a 4-cycle

$$(a, b, a), (c, b, a), (c, b, c), (a, b, c), \qquad \{a, b, c\} = \{0, 1, 2\}.$$

The three disjoint forced cycles prevent a Hamilton cycle. For the Hamilton-connected lower bound, $\mathcal{R}_3(K_2) = C_6$ is already not Hamilton-connected. □

5 Hamiltonian four-colour subclasses at girth five

The reductions below splice Hamilton cycles and establish Hamiltonicity at four colours. They do not prescribe arbitrary endpoints, so the Hamilton-connected thresholds of these subclasses remain in $[4, 5]$. Supplementary Appendices B–G supply their local certificates.

Call a vertex *heavy* if its degree is at least four, and a pentagon *ordinary* if all five of its vertices have degree three. These degrees are measured in the graph before deletion.

5.1 Graphs with few five-cycles

Lemma 5.1. *Let $S = \{u_1, \ldots, u_s\}$ be independent, $s \geq 3$, and suppose $N(u_i) = \{w, x_i\}$, where $N(w) = S$, $w \notin S$, and the x_i are distinct and belong to $J = G - (S \cup \{w\})$. If $\mathcal{R}_4(J)$ is Hamiltonian and some proper colouring of J gives the same colour to two distinct x_i, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. Take a Hamilton cycle C of $\mathcal{R}_4(J)$. Write three distinct chosen edges as $e_r = f_r g_r$ $(r = 1, 2, 3)$. Since one edge of C changes only one vertex of J and the x_i are distinct, choose $j(r)$ so that $x_{j(r)}$ is unchanged on e_r, and put $a_r = f_r(x_{j(r)}) = g_r(x_{j(r)})$.

There is a tree T on the four colours whose edge t_r contains a_r. Indeed, if the three labels are distinct, use the star centred at the fourth colour; if exactly two labels are equal, say a, a, b, use the edges ab, ac, bd, where c, d are the two remaining colours; and if all labels are equal, use the star centred at that common label. Make four copies of C, indexed by the colour of w. For each

r, remove the copy of e_r from the two copies indexed by the ends of t_r, and add the two cross edges joining corresponding J-colourings while changing the colour of w. These are legal edges of $\mathcal{R}_4(G-S)$ because w is isolated in $G-S$. The three two-edge switches join the four cycles according to the tree T, so the resulting 2-regular connected graph is a Hamilton cycle B of $\mathcal{R}_4(G-S)$.

The only edges of B that change the colour of w are these cross edges. At the one indexed by r, the unchanged vertex $x_{j(r)}$ has colour a_r, and a_r is one of the two endpoint colours of t_r. Thus every such transition has an unchanged x_j whose colour is one of the two colours of w; this is the required control property.

Over a base colouring f the u_i-lists are

$$A_i(f) = [4] \setminus \{f(w), f(x_i)\}.$$

They have size at least two. At a move in J, one list changes and another coordinate is unchanged. At a move of w, all intersections are $[4] \setminus \{a, b, f(x_i)\}$, and the control property leaves a coordinate with a two-element intersection. Thus every interface has an edge, and the cyclic product-fibre lift applies once we exhibit its nonbinary fibre and the two distinct incident interface edges. Let h be the assumed colouring of J with $h(x_p) = h(x_q) = c$, and take the base state with restriction h and w coloured c. The two lists at u_p, u_q then both have size three. If an incident edge of B changes a vertex of J, at most one of these two lists changes, so the other unchanged list supplies a K_3 in the common interface. If it changes w from c to b, both common lists contain $[4] \setminus \{c, b\}$ and hence supply the two distinct edges of $K_2 \square K_2$. Consequently the two incident interfaces contain at least two distinct edges, from which distinct edges can be chosen for the cyclic lift. Its fibre over the displayed base state has a factor K_3; the lift therefore gives a Hamilton cycle through every extension of every base state, proving the lemma. □

Lemma 5.2. *Let C be a connected plane graph of girth at least five and minimum degree at least two. Let p be the number of 5-faces, $Q = \sum_{|F|\geq 7}(|F| - 6)$, $s = |\{v : d(v) = 2\}|$, and $D = \sum_v (d(v) - 3)$ over vertices of degree at least three. Then*

$$p = 12 + Q + 2D - 2s.$$

Every 5-face is a simple 5-cycle, and distinct faces give distinct cycles unless $C = C_5$.

Proof. A facial walk has no immediate reversal and contains a cycle, hence has length at least five. Counting face lengths gives $2e = 6f - p + Q$, while the degree sum is $2e = 3n - s + D$; with $n - e + f = 2$, subtraction yields the identity. A repeated vertex on a facial walk of length five would give a shorter cycle. If one 5-cycle bounds two faces, all its vertices have degree two and connectedness gives $C = C_5$. □

Corollary 5.3. *A planar graph of girth at least five with at most eleven distinct 5-cycles is two-degenerate.*

Proof. Otherwise a connected subgraph has minimum degree at least three. In its plane embedding the identity has $s = 0$, and therefore at least twelve 5-faces, which are distinct 5-cycles. This contradicts the hypothesis. □

Theorem 5.4. *Let G be a nonempty planar graph of girth at least five with at most fifteen distinct 5-cycles. Then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. The uniform Hamiltonian threshold of this class is four.*

Proof. The class is closed under vertex deletion. Choose a minimum four-colour counterexample. By Lemma 5.16, it is connected and has minimum degree at least two. The case C_5 is covered by Theorem 5.27. Fix a plane embedding and use the notation of (5.1), so $p \le 15$.

If $s = 0$, that identity gives $D \le 1$, so there is at most one heavy vertex, contrary to Theorem 5.27. If $s = 1$, Lemma 5.17 gives $p \ge 20$. Thus $s \ge 2$, with a common heavy neighbour w and $t \ge 1$. The inequalities $D' \ge h' \ge 3$ and $p \le 15$, together with (5.2)–(5.3), force

$$t = 1, \qquad D' = h' = 3, \qquad Q = R = 0, \qquad p = 14.$$

All three other heavy vertices have degree four. By (5.5), all their face incidences are pentagonal, and no pentagonal face contains two of them.

Since $d(w) = s + 1 \ge 4$, there is an angle at w between two edges to S. Its face has a simple boundary of length five or six by Lemma 5.18. Removing w from this boundary gives a shared-hub path whose endpoints have degree two and whose internal vertices lie outside S. If the face is pentagonal, at most one internal vertex is heavy, of degree four. Lemmas 5.12 and 5.13 exclude the degree sequences $(2, 3, 3, 2)$ and $(2, 4, 3, 2)$, allowing reversal. If the face is hexagonal, no internal vertex is heavy, since every other-heavy incidence is pentagonal. The order-five case of Lemma 5.12 applies. Either case contradicts minimality.

For $k \ge 5$, Theorem 4.5 applies. The frozen periodic three-colouring 012012 of C_6, which has no 5-cycle, gives the lower bound four. □

5.2 Local degree-two reductions

Lemma 5.5. *Let u, v be adjacent degree-two vertices, with $N(u) = \{v, x\}$ and $N(v) = \{u, y\}$, where $x \ne y$. If $\mathcal{R}_4(G - \{u, v\})$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. For a base colouring f, the extension fibre is $F(A, B)$, where $A = [4] \setminus \{f(x)\}$ and $B = [4] \setminus \{f(y)\}$. For $A = B = \{a, b, c\}$ use the cycle (ab, ac, bc, ba, ca, cb); for $A = \{a, b, c\}$, $B = \{a, b, d\}$ use $(ab, ad, bd, ba, ca, cd, cb)$.

In fact, any two fibre Hamilton cycles have at least two common edges when at most one list changes. For an unchanged fibre, equal lists give C_6, while unequal lists give four distinct forced edges at the two degree-two states using the common colours. By exchanging coordinates suppose only A changes. If one value equals $B = \{a, b, c\}$, the common fibre is the path (ab, ac, bc, ba); its two end edges are forced in both cycles. Otherwise let a, b be the old and new excluded colours, c the fixed excluded colour and d the fourth colour. The common fibre has five vertices ca, cb, cd, da, db and six edges. The old outside states ba, bd occur consecutively because bd has degree two with neighbours ba, cd. Thus the old cycle uses exactly four common-fibre edges; the new cycle does likewise with ab, ad, cd. The cycles share at least $4 + 4 - 6 = 2$ edges.

Since $x \ne y$, a base step changes at most one list. Cut an edge of a base Hamilton cycle and process its fibres in path order. At each splice the current fibre has lost at most one edge and the next is fresh, so a common edge remains. Cutting its two copies and adding the two actual base-step edges merges the circles. The fibres partition all states, giving a spanning cycle. □

Lemma 5.6. *Let G be planar of girth at least five and let u, v be nonadjacent vertices of degree two with $N(u) \cap N(v) = \varnothing$. If $H = G - \{u, v\}$ has Hamiltonian $\mathcal{R}_4(H)$, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. Set $S = \{u, v\}$. Every vertex of H has at most one neighbour in S. Let x, y be the neighbours of u; they are distinct and nonadjacent. The planar triangle-free graph $H - \{x, y\}$ admits a proper three-colouring by Grötzsch's theorem [4] (the empty graph has its empty colouring). Use colours $1, 2, 3$ there and give both x, y colour 4. This is a proper colouring of H leaving three colours

available at u. The hypotheses of Lemma 3.5 hold with $k = 4$ and $|S| = 2$, so that lemma supplies the required Hamilton cycle. □

5.3 Restoring a $(2, 3, 2)$ path

The following reduction uses only a Hamilton cycle of the deleted graph. It processes an entire constant-boundary run together with its two neighbouring nonconstant fibres, retaining the edges needed for the next splice. The three external neighbours remain distinct actual vertices.

Proposition 5.7. *If $F = \mathcal{R}_3(P_3)$, then $F \,\square\, P_\ell$ is Hamiltonian for every $\ell \geq 2$.*

Proof. Partition F into its three centre-colour squares Q_b. Order each square as (q_0, q_1, q_2, q_3), where q_0, q_2 are its monochromatic-endpoint vertices. On $Q_b \square P_2$ use

$$(0, q_0), (1, q_0), (1, q_1), (0, q_1), (0, q_2), (1, q_2), (1, q_3), (0, q_3).$$

This cube cycle contains all four edges between layers 0 and 1. For larger ℓ, append each subsequent square by an ordinary edge splice using an internal edge of the newest layer. The induction invariant is that the strip is one cycle, contains every vertex in the first ℓ layers, retains all four edges between layers 0 and 1, and has at least two unused internal edges in its newest layer. The initial strip has two such edges; after one splice the new layer has three, so a further splice can avoid any prescribed reserved edge. Thus the invariant holds for every $\ell \geq 2$, and the four original interlayer edges are never cut.

The three centre-colour squares of F are connected by three edges between monochromatic-endpoint vertices, one for each pair of centre colours. Choose a tree of two of these edges. Each chosen edge xy produces a square with vertices $(0, x), (1, x), (1, y), (0, y)$. Delete its two interlayer edges and add its two copies of xy, joining the corresponding strip cycles. The two selected interlayer edges at the middle strip are distinct. Two such splices give a single spanning cycle, for arbitrary $\ell \geq 2$. □

Let x, y, z be a path of degrees $2, 3, 2$. Its external neighbors, in that order, are p, q, r. Assume they are distinct and outside the path. For a bottom colouring with boundary $t = (f(p), f(q), f(r))$, write

$$B_t = \{(u, v, w) : u \neq v,\ v \neq w,\ u \neq t_0,\ v \neq t_1,\ w \neq t_2\}, \qquad u, v, w \in \{0, 1, 2, 3\}.$$

A boundary is *bad* if all three entries agree, and *good* otherwise. Up to colour permutations and path reversal the good types are $001, 010, 012$. Their orders are $6 + 4 + 4 = 14$, $9 + 4 + 4 = 17$, and $6 + 6 + 4 = 16$, respectively. The bad type 000 has order 12.

Good fibres and safe edges. The cycles D_E, D_M, D_A for the normalized boundaries $001, 010, 012$ are specified by literal cycles and capacity checks in supplementary Appendix B; the coordinates are uvw. Allow all their colour images and path reversals. Legality, cyclic adjacency, distinctness, and the counts above verify that each is a Hamilton cycle of its fibre.

For a bad boundary aaa, partition B_{aaa} into squares Q_c indexed by $c \neq a$, with fixed middle coordinate c. A good neighbour of aaa is obtained by changing position j from a to $b \neq a$. If j is an end position, then for each $c \notin \{a, b\}$ there is exactly one edge common to the good fibre and Q_c. Let P_t consist of these two edges. For middle-change or all-distinct boundaries set $P_t = \varnothing$. A good-cycle edge outside P_t is called *safe*. Directly from the supplementary path certificate, every allowed cycle has the following properties:

1. For an end change, it contains the unique edge in each Q_c, $c \notin \{a, b\}$, and two opposite edges of Q_b.

2. For a middle change it has no vertex in Q_b and contains three edges of each of the other two squares.

For $t = 001$, the two special edges are 123–323 and 132–232; the two edges in Q_1 are 313–213 and 212–312. For $t = 010$, the retained three-edge paths are $121, 321, 323, 123$ and $132, 232, 231, 131$. These descriptions also prove the assertions for every symmetry image.

Lemma 5.8. *For adjacent good boundaries t, s, every allowed cycle at t and every allowed cycle at s have at least two common edges outside $P_t \cup P_s$.*

Proof. Partition the common fibre graph into its fixed-middle layers H_v and the edge set H_* changing the middle coordinate. In any block, if m edges remain outside $P_t \cup P_s$ and the two cycles retain x, y of them, they share at least $\max(0, x + y - m)$ safe edges. These bounds add over disjoint blocks. The supplementary path-kernel certificate classifies all allowed transitions and gives sufficient block counts for seven families, each yielding at least two.

Let a, b, c, d be distinct and use the selected cycles defined there. For $aab \to bab$, the common middle-b prism has no special edges. The E cycle retains one edge in each row triangle and two vertical edges. The cycle M_0 contains both these triangle edges, and M_1 both these vertical edges, giving two safe common edges in either case.

For $aba \to cba$, H_a has nine edges. Both cycles omit *bab–bac* and *bab–bad*: the M layer isolates *bab*, while the A path starts bab, dab. They retain three and five edges in this seven-edge universe, giving one common edge. In H_c they retain three and two of four edges, giving another. Both special sets are empty. Colour permutations and reversal transport all arguments, proving the assertion for every allowed labelled pair. □

Bad blocks.

Lemma 5.9. *Consider consecutive fibres with boundaries*

$$t, \underbrace{aaa, \ldots, aaa}_{\ell \text{ times}}, s, \qquad \ell \geq 1,$$

where t, s are good and each differs from aaa in one coordinate. When $\ell = 1$ suppose $t \neq s$. Choose any allowed good cycles at the two ends, and reserve at most one safe edge in each. The layered graph on these fibres has a Hamilton cycle retaining both reserved edges. Only consecutive-layer matching edges are used.

Proof. Call a common edge with Q_c *available* if it belongs to the chosen good cycle and is not its reserved edge. An end-change good cycle has available edges in all three squares: the two singleton edges are special, hence cannot have been reserved, and at least one of its two Q_b edges remains. A middle-change good cycle has at least two available edges in each of its two supported squares.

First suppose $\ell = 1$. Let S_L, S_R be the sets of square colours supported by available edges of the two chosen good cycles. Then $|S_L|, |S_R| \geq 2$ and $S_L \cup S_R = [4] \setminus \{a\}$: the only exception could be two middle changes to the same colour, which would give $t = s$.

Choose a common square with different available edges on the two sides. If either change is at the middle, any $c \in S_L \cap S_R$ works, since that side offers at least two edges. If both changes are at ends, choose $c \notin \{a, b_L, b_R\}$, where b_L, b_R are their changed colours. Its two singleton edges are

different: different ends give different coordinate directions, while changes at the same end have $b_L \neq b_R$ and give opposite square edges.

Start with the two good cycles and three bad squares. Attach the chosen square to both good cycles using these different edges, and attach each other square to a supporting good cycle. These four incidences form a tree. Cuts at a good cycle lie in different middle-colour squares, and the two bridge-square cuts are distinct. Along each tree edge delete the two selected internal edges and add the two matching edges. Each splice joins disjoint cycles into one; all reserved edges survive.

Now let $\ell \geq 2$. Each good end supports at least two squares, so choose an available left edge in Q_c and an available right edge in Q_d with $c \neq d$. Build the three square-strip cycles as in Proposition 5.7. The cube formula there permits either perfect matching of a square to be used in the first layer, by rotating its cyclic order; the last layer of a two-layer strip uses the other matching. Consequently its first layer can retain any specified square edge. For a two-layer strip the same is true at its last layer; our two requirements are on different strips and are independent.

If $\ell \geq 3$, a last-layer requirement is met by appending each new square with a splice edge different from the required edge. There are initially two choices and subsequently three in the current last layer, so this is always possible. The new last square retains all edges except the splice edge. First-layer horizontal edges and all four original vertical edges survive these extensions.

Join the three strips using the two vertical-edge switches in Proposition 5.7; these do not affect either required horizontal edge. Finally join the left and right good cycles to their selected strip edges by two ordinary edge splices. The result is one Hamilton cycle. The reserved edges were never cut. This proves the claim for every length, rather than just a bounded set of runs. □

Restoration over a base cycle.

Lemma 5.10. *Suppose the three external neighbours of a degree-*$(2, 3, 2)$ *path are distinct and outside it. If the reconfiguration graph of the deleted graph is Hamiltonian at four colours, then so is* $\mathcal{R}_4(G)$.

Proof. Fix an arbitrary bottom Hamilton cycle. There must be a good vertex on it. Otherwise every boundary is constant; consecutive constant triples can only be equal, as at most one coordinate changes. The external colour would therefore be fixed throughout the cycle. A global colour permutation of a proper colouring supplies another proper colouring with a different external colour, a contradiction to Hamilton coverage. Thus even the startup uses only Hamiltonicity.

Every good bottom vertex has at most one bad neighbour in the entire bottom reconfiguration graph: a two-colour boundary has exactly one possible coordinate change making it constant, and an all-distinct boundary has none. The full neighbouring colouring, if proper, is uniquely determined by that change. Hence no good vertex on the Hamilton cycle can lie between two bad vertices. In particular there is an edge joining two good vertices.

Cut such an edge. In the resulting bottom Hamilton path, group each maximal bad run together with its two neighbouring good vertices; make each remaining good vertex a singleton block. These blocks are disjoint and partition the path. A bad run has constant boundary, and its two neighbours are distinct. If the run has length one their boundaries are different: otherwise both neighbours are obtained from the intervening colouring by the same one-vertex recolouring and would be the same bottom vertex.

For each labelled good boundary fix one allowed cycle once and for all. Consecutive blocks meet at a good–good boundary transition. If the boundary changes, Lemma 5.8 provides two common safe edges. If it stays unchanged, the fixed cycles coincide and have at least $14 - 2$ safe edges. Select one connecting edge for each interblock interface in path order. If a singleton block already has an

incoming selected edge, choose a different outgoing one, which is possible from the two candidates. In a nonsingleton block the two selected edges are in its distinct end fibres, so cannot conflict.

Use Lemma 5.9 to construct each nonsingleton block cycle preserving its selected end edges; singleton blocks already have their fixed cycles. Splice the resulting disjoint block cycles along the selected interfaces. Each step joins two cycles into one, and every later selected edge is still present. The one-block case needs no final splice. All transitions are legal: a local step changes one path vertex, and a matching cross-edge performs its actual bottom recolouring while holding the local colouring fixed. Finally the fibres are disjoint and exhaust all proper colourings. Thus the resulting single cycle is Hamiltonian, including its closing edge. □

5.4 Paths with a common outside neighbour

For $r \in \{4, 5\}$ and $t = (a, b_1, \ldots, b_{r-2}) \in Q^{r-1}$, where $Q = \{0, 1, 2, 3\}$, let $F_t^{(r)}$ consist of the proper colourings of the path $v_0 \cdots v_{r-1}$ that avoid the respective colours $(a, b_1, \ldots, b_{r-2}, a)$. Two words are adjacent when they differ in one coordinate. The palette Q is a relabelling of [4]; the first boundary coordinate is the colour of one outside neighbour shared by the endpoints.

Lemma 5.11. *There are Hamilton cycles $C_t^{(r)}$ in all $F_t^{(r)}$ such that whenever t, t' differ in one coordinate, the chosen cycles have at least two common edges.*

Proof. Supplementary Appendix E specifies five and fifteen literal cycle templates for $r = 4$ and $r = 5$, respectively. Normalize colours in first-appearance order, append unused colours in increasing order, and restore each word by the inverse palette map. The exact reader checks coverage and cyclic adjacency for all 64 and 256 labelled boundaries, then intersects the chosen cycle edge sets at all 576 and 3,072 directed one-coordinate interfaces. The minimum intersections have three and five edges, respectively. In particular, changing a changes both endpoint exclusions in one step. □

Lemma 5.12. *Let G have girth at least five. Suppose $v_0 \cdots v_{r-1}$ is a path, where $r \in \{4, 5\}$, its endpoints have degree two and a common neighbour w outside the path, and its internal vertices have degree three. Put $H = G - \{v_0, \ldots, v_{r-1}\}$. If $\mathcal{R}_4(H)$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.*

Proof. There are no additional internal edges. The third neighbours of the internal vertices are distinct and different from w: a prohibited coincidence would create a triangle or a 4-cycle. The extension fibres are therefore exactly the graphs in Lemma 5.11.

Cut an edge of an arbitrary bottom Hamilton cycle and process its fibres in path order, giving each its certified cycle. An outside move either leaves the boundary unchanged or changes one physical coordinate. The chosen cycles share at least two internal edges. Both endpoints of a common edge are legal in both fibres, so the two matching cross edges perform the actual outside recolouring.

The current last fibre has lost at most its one incoming splice edge. Choose a remaining common edge, remove it from the current cycle and the fresh fibre cycle, and add the two matching cross edges. This joins the cycles. Iteration gives one cycle through every extension of every bottom colouring, and hence a Hamilton cycle of $\mathcal{R}_4(G)$. □

5.5 Further degree-two reductions

A shared-hub path has degree-two endpoints with the same neighbour outside the path. For $m \in \{5, 6\}$, a tailed m-cycle consists of an induced cycle $v_0 \cdots v_{m-1} v_0$ and an additional vertex u whose only neighbour on the cycle is v_0, where

$$d(v_0) = 4, \qquad d(v_1) = d(u) = 2, \qquad d(v_i) = 3 \quad (2 \le i < m).$$

The other neighbour of u is outside this configuration. All degrees in the following lemma are degrees in G.

Lemma 5.13. *Let G have girth at least five, and let X be one of the following configurations:*

1. *a degree-four vertex and three of its degree-two neighbours;*
2. *an induced shared-hub path with degree sequence $(2,4,3,2)$, $(2,4,3,3,2)$, $(2,3,4,3,2)$, or $(2,3,3,3,3,2)$, allowing reversal;*
3. *a tailed 5-cycle or tailed 6-cycle.*

If $\mathcal{R}_4(G-X)$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.

Proof. Supplementary Appendix F specifies seven literal cycle families for these configurations. A port denotes an outside-neighbour role; for a shared-hub path, one port controls both endpoints. Colours are normalized by first appearance, with unused colours appended in increasing order. The inverse palette map selects one fixed cycle at every labelled boundary.

Two distinct port roles can denote the same outside vertex only if all pairs of their local neighbours have internal distance at least three; otherwise a triangle or a 4-cycle results. Appendix F lists all allowed pairs. Their compatibility graphs have no triangle, so no three distinct roles can coincide. Several disjoint pairs may coincide, but an actual outside move changes only one singleton or one listed pair. Both entries of a pair change together from the same old colour to the same new colour.

The exact reader reconstructs the seven fixed internal graphs and all legal states, checks the 421 normalized cycles, and verifies all 8,704 labelled boundaries and 158,976 directed interfaces, including the paired changes. The chosen cycles share at least two edges at every interface. A separate reader restores every cycle into physical colour labels and directly checks the same interface inequalities. Table F.1 gives the individual counts.

Cut an edge of a bottom Hamilton cycle and process its fibres in path order. Consecutive fibres have at least two common chosen-cycle edges; an unchanged boundary has identical chosen cycles. The current last fibre has lost at most its incoming splice edge. Choose another common edge, remove its two copies, and add the two matching cross edges, which perform the actual outside recolouring with the local state fixed. This joins two cycles and preserves the invariant. Iteration gives one cycle through all extensions of all bottom colourings, as required. □

5.6 Pentagonal reservoirs

Lemma 5.14. *With palette $Q=\{0,1,2,3\}$, the following models admit fixed Hamilton cycles in every component of every boundary fibre. The last column lists the outside-neighbour roles whose colours each local vertex must avoid; cycle vertices are ordered $v_0,\dots,v_4$, followed by z when present.*

model	*internal graph*	*forbidden role sets*
R	$C_5 \sqcup \{z\}$	$\{0\},\{1\},\{2\},\{3\},\{4\},\{5,6\}$
F	C_5	$\varnothing,\{0\},\{1\},\{2\},\{3\}$
P	C_5+v_0z	$\varnothing,\{0\},\{1\},\{2\},\{3\},\{4\}$

Every fibre of F and P is connected. A fibre of R has two components exactly when its first five boundary entries are equal, and is otherwise connected. Across every admissible physical boundary move, each component on either side overlaps a component on the other side. For every overlapping pair, the selected component cycles have at least four common edges in R, and at least two in F and P.

Proof. Supplementary Appendix G specifies the three literal cycle families. Normalize colours by first appearance, appending unused colours in increasing order, and restore words by the inverse palette map. This selects the cycles simultaneously at all labelled boundaries. The exact readers reconstruct the legal states and components, check cyclic coverage and adjacency, and test all labelled interfaces. A separate physical-label reader reproduces the interface histograms.

The three families have respectively $715, 15, 51$ normalized boundaries, $720, 15, 51$ component cycles, and 16,384, 256, 1,024 labelled boundaries. The numbers of directed boundary interfaces are 466,944, 3,072, 16,896, with minimum common-edge counts $4, 11, 8$. For R, all 470,208 overlapping component interfaces are checked.

In R, the first five roles are distinct, as are the last two. The possible extra identifications are the pairs (i, j) with $0 \leq i < 5$ and $j \in \{5, 6\}$. Their simultaneous identifications form the 31 matchings of this bipartite graph. In F all roles are distinct. In P the only possible pairs are $(1, 4)$ and $(2, 4)$, and they cannot both coincide. Girth excludes every other identification. Checking all labelled single-role and simultaneous equal-pair moves covers these possibilities, including other static identifications in the same boundary. □

Lemma 5.15. *Let G have girth at least five, and let X be one of the following:*

1. *an ordinary pentagon and a degree-two vertex outside it with no neighbour on the pentagon;*
2. *a pentagon with one degree-two vertex and all other vertices of degree three;*
3. *an ordinary pentagon and an adjacent degree-two vertex outside it.*

If $\mathcal{R}_4(G - X)$ is Hamiltonian, then $\mathcal{R}_4(G)$ is Hamiltonian.

Proof. Girth excludes chords and gives exactly the three models of Lemma 5.14. In the last case the outside degree-two vertex meets the pentagon only once; deleting it leaves a full list at that cycle vertex. Every matching cross edge at a common local state performs the actual bottom recolouring, since that state is legal in both fibres.

Cut a bottom Hamilton cycle into a path and give each component of each fibre its fixed selected cycle. For the two connected-fibre models, splice successive cycles in path order. The current last fibre has lost at most one incoming splice edge. Of at least two common cycle edges with the next fresh fibre, one remains; deleting its two copies and adding the matching cross edges joins the cycles. An unchanged boundary uses identical cycles.

For R, form a graph whose nodes are the components of the layers, joining overlapping components in consecutive layers. It has maximum degree at most four. Some layer is connected. Otherwise the five cycle-neighbour colours form a constant word at every bottom state. A physical move changes at most one of these five entries, so that constant word is unchanged along the bottom path. A palette permutation changing its colour produces another proper bottom colouring, contradicting Hamilton coverage.

Every component can step to a component in either adjacent layer. Walking towards a connected layer therefore proves that the component graph is connected. Choose a rooted spanning tree and attach each fresh component cycle to its already attached parent. Before an attachment the parent has lost at most three internal edges, since its tree degree is at most four and one attachment remains. At least four common selected-cycle edges are available, so an uncut edge remains for the switch. An unchanged boundary uses identical component cycles, which also have at least four edges. Each switch joins two cycles. The final cycle covers every component of every fibre, hence all proper colourings of G. □

5.7 Degree-two vertices in a minimum counterexample

In the next two lemmas, a minimum counterexample means a graph of minimum order with non-Hamiltonian $\mathcal{R}_4(G)$ in a fixed class of nonempty planar graphs of girth at least five closed under nonempty vertex deletion. Thus every smaller nonempty induced graph has a bottom Hamilton cycle.

Lemma 5.16. *A minimum counterexample is connected and has minimum degree at least two. Let S be its set of degree-two vertices. If $|S| \geq 2$, its members have a common heavy neighbour. If $S \neq \varnothing$, every 5-cycle meets a heavy vertex.*

Proof. The singleton case gives K_4. Products of Hamiltonian graphs are Hamiltonian, by alternating two distinct splice edges along a factor Hamilton path, so G is connected. Lemma 2.2 excludes degree at most one, and Lemma 5.5 makes S independent.

Suppose $|S| \geq 2$. If no outside vertex is adjacent to all of S, every outside vertex has at most $|S| - 1$ neighbours in S. Put $H = G - S$ and choose $u \in S$ with neighbours x, y. They are nonadjacent by girth. Three-colour $H - \{x, y\}$ by Grötzsch's theorem [4], and give both x, y colour four. This leaves three colours available at u, so Lemma 3.5 contradicts minimality. There is consequently a common neighbour w outside S. It cannot have degree two, since S is independent, and degree three would give Lemma 5.10; the three external neighbours of that path are distinct by girth. Thus w is heavy. Write $N(u_i) = \{w, x_i\}$; the x_i are distinct because a repetition creates a 4-cycle.

If $S = \{z\}$, a pentagon avoiding heavy vertices either contains z and has all other degrees three, or is ordinary. Pair an ordinary pentagon with z, using its adjacency or nonadjacency to the cycle. All three possibilities are excluded by Lemma 5.15. If $|S| \geq 2$, a cycle containing a member of S must contain w. A pentagon avoiding heavy vertices is therefore ordinary; pairing it with any member of S gives the same contradiction. □

For a plane minimum counterexample write

$$s = |S|, \qquad h = |\{v : d(v) \geq 4\}|, \qquad D = \sum_{d(v)\geq 4} (d(v) - 3), \qquad Q = \sum_{|F|\geq 7} (|F| - 6),$$

and let p count pentagonal faces. Lemma 5.2 gives

$$p = 12 + Q + 2D - 2s. \tag{5.1}$$

Pentagonal faces give distinct 5-cycles unless $G = C_5$, already covered by Theorem 5.19.

Lemma 5.17. *If a minimum counterexample has $s = 1$, then*

$$10 + D + Q \leq 3h, \qquad h \geq 5, \qquad p \geq 20.$$

If $s \geq 2$, write

$$N(u_i) = \{w, x_i\}, \qquad d(w) = s + t,$$

and let h' count the heavy vertices other than w, with $D' = \sum_{v\neq w,\ d(v)\geq 4}(d(v) - 3)$. Then $t \geq 1$ and

$$p = 6 + Q + 2t + 2D', \tag{5.2}$$

$$D' + 6 + Q \leq 3h', \qquad D' \geq h' \geq 3. \tag{5.3}$$

Proof. If $s = 1$, every pentagonal face meets a heavy vertex by Lemma 5.16. Counting its incidences gives $p \le 3h + D$. Combine this with (5.1) and $D \ge h$ to obtain all three assertions.

For $s \ge 2$, substituting $D = s + t - 3 + D'$ gives (5.2). A pentagonal face having no heavy vertex other than possibly w must contain w. It cannot use two edges from w to S: its other two cycle vertices would give the regular shared-hub path of order four in Lemma 5.12. Thus it uses an edge from w to $Z = N(w) \setminus S$, so there are at most $2t$ such faces. Counting all remaining pentagonal faces at other heavy vertices gives $p \le 2t + 3h' + D'$. This proves (5.3), since $D' \ge h'$.

If $t = 0$, then $s = d(w) \ge 4$. Among the distinct x_i, two are nonadjacent, since otherwise three form a triangle. Put $J = G - (S \cup \{w\})$. Three-colour J with these two vertices removed by Grötzsch's theorem, then give both colour four. Induction on J and Lemma 5.1 contradict minimality. Therefore $t \ge 1$. □

For $s \ge 2$, retain the nonnegative deficit

$$R = 2t + 3h' + D' - p = 3h' - D' - 6 - Q \ge 0. \tag{5.4}$$

Let p_0 count pentagonal faces with no heavy vertex other than possibly w, and let $a(F)$ count incidences of heavy vertices other than w on a face F. Summing these incidences gives the exact decomposition

$$R = (2t - p_0) + \sum_{|F| \ne 5} a(F) + \sum_{|F|=5,\ a(F)>0} (a(F) - 1). \tag{5.5}$$

Every term is nonnegative. Incidences count repeated occurrences on a nonpentagonal facial walk.

Lemma 5.18. *In a connected plane graph of girth at least five and minimum degree at least two, a nonsimple facial walk has length at least ten. Put $Q = \sum_{|F| \ge 7} (|F| - 6)$. If $Q \le 1$, every face has a simple boundary of length at most seven, and distinct angles at any vertex belong to distinct faces.*

Proof. A facial walk is cyclically nonbacktracking. Splitting it at a repeated vertex gives two nonempty closed subwalks without internal immediate reversal. Each contains a simple cycle: a nontrivial closed walk in a tree reverses at an internal leaf. Both subwalks therefore have length at least five. If $Q \le 1$, no face has length at least eight, so every boundary is simple. A simple boundary visits a vertex at most once, proving the last assertion. □

5.8 Protected pentagon restoration

Theorem 5.19. *Let G be a nonempty finite simple planar graph of girth at least five and maximum degree at most three. Then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \ge 4$. Disconnected graphs are allowed. The uniform Hamiltonian threshold of this class is exactly four, and its Hamilton-connected threshold lies between four and five.*

We prove a local restoration lemma using only Hamiltonicity of the smaller recolouring graph, and then supply the planar structure. Supplementary Appendices C–D specify the cycles, seeds, pools and interface records for 2,304 labelled pentagon boundaries and 313 standard-source one-entry interfaces. We first prove the protected-edge invariant, then use it in an arbitrary-order restoration.

Local certificates. Put $Q = \{0, 1, 2, 3\}$. A boundary t has five entries in Q, with at most one entry $*$. Its local graph F_t consists of the proper C_5 colourings x satisfying $x_i \ne t_i$; $*$ excludes no colour. In the certificates $*$ is encoded by 4. The eleven ordinary types have 590 states in total. The constant type has two components, each a 15-cycle; all other ordinary types are connected. Eight

have the printed Hamilton cycles, and the two remaining types are 00001 and 00112. With a full list, the eleven types, after rotation, reflection and colour permutation, are

$$0000*,\ 0001*,\ 0010*,\ 0011*,\ 0012*,\ 0101*,\ 0102*,\ 0110*,\ 0112*,\ 0120*,\ 0123*.$$

Their printed Hamilton cycles have respectively $54, 60, 78, 62, 71, 88, 85, 74, 74, 83, 82$ states, a total of 811. Thus there are 23 component types, including the two constant components.

We record the four local facts used in the restoration as Claims 1–4.

Claim 1. *Every one of the 23 component types B has a partition into paths $P_1, \dots, P_r$ of order two or three, with an actual B edge joining P_i to P_{i+1} for each $i < r$.*

Proof. Twenty-one types have the printed Hamilton cycles, and 00112 has the 51-state Hamilton path in supplementary Appendix C. Delete a cycle edge where necessary, and divide the resulting path into consecutive two-vertex blocks, with one three-vertex block if its order is odd. For 00001, supplementary Appendix C supplies the ordered path blocks and their actual connectors. The verifier checks coverage and every one-coordinate difference. □

For a Hamilton component A with specified cycle C_A, designate an ordered pool of five edges for either constant component, and eight edges otherwise. Let K_A be the other cycle edges. The first two pool edges join copies of C_A in a two- or three-row path product; the remaining pool edges supply at least two splice reserves, all outside K_A. For a hard component, specify cycles of $A\square P_2$ and $A\square P_3$, and a set D_A of internal edges occurring in *every* row of both seeds. For each of 00001 and 00112, supplementary Appendices C–D specify a two-row cycle and derive its three-row seed by the same three-path insertion rule. Both use $|D_A| = 13$ and a two-edge pool. Put $K_A = D_A \setminus \text{pool}$; the two distinct pool edges supply the splice reserves.

Here is the exact normalization convention. Enumerate $s = 0, \dots, 4$, then $\epsilon = +1, -1$, then the permutations $p \in S_4$ in lexicographic order. Take the first standard type $n_i = p(t_{s+\epsilon i})$, retaining $*$. Restoration puts the standard entry x_i at position $s + \epsilon i$ with colour $p^{-1}(x_i)$. In particular a standard boundary first normalizes by the identity.

Claim 2. *Let the source be a standard component A of F_t, and change one non-star boundary entry. Normalize the target by the preceding rule. Every target component B whose restored state set meets A has an edge in common with K_A after restoration of K_B. For an unchanged boundary use the same component and the same map.*

Proof. The ordinary and starred orbit representatives cover $4^5 + 5 \cdot 4^4 = 2304$ labelled boundaries. The displayed cyclic words give their specified component cycles and the two-row hard seeds: their entries are precisely the legal words, without repetition, and successive entries, including last and first, differ in one coordinate. Both three-row seeds follow from the supplement's three-path insertion rule, using two edges and the displayed 40-state or 47-state path. Every listed common edge occurs in every row of both corresponding seeds. Removing the listed pools therefore defines the asserted protected sets. For the 313 standard-source interfaces, supplementary Appendix D gives 47 individual common edges, twenty-one source-selector rows covering 236 cases, and six constant-source families covering 30 cases. Endpoint membership in the specified templates and outside their pools proves each individual record. The selector rows use fixed protected incident edges against the opposite component's full degree, with the prescribed first normalizer and pools. The constant-source families use two normalizing directions and four protected target edges. This is a complete finite case proof for standard sources; labelled sources are obtained by transport, with the target normalized relative to the source as described after Lemma 5.21. □

Products and overlaps.

Lemma 5.20. *For any two of the 23 component types A, B, the graph $A\square B$ has a Hamilton cycle retaining a copy of every edge of K_A in every physical B row.*

Proof. Use the ordered path blocks $P_1, \dots, P_r$ of Claim 1 for B. For a two-row block with rows b_0, b_1, remove the same pool edge xy from the two copies of C_A and add

$$(x, b_0)(x, b_1), \qquad (y, b_0)(y, b_1).$$

This is one cycle on $A\square P_2$. For a three-row block use two distinct pool edges x_1y_1, x_2y_2 and perform this operation between rows b_0, b_1 and b_1, b_2, respectively. The two cuts in the middle row are distinct, so the result is again one cycle. For either hard component, use the literal two- or three-row seed; the certificate checks that every edge of D_A occurs in every physical row of that seed. In all cases every edge of K_A survives in every row. For an ordinary component retain two of the unused pool edges as reserves; for a hard component the two pool edges in its seed are the reserves. Thus every reserve is outside K_A.

Fix two such reserves e and f. Between P_i and P_{i+1}, let $u_i \in P_i$ and $v_i \in P_{i+1}$ be the certified adjacent states. Cut the copies of e in the two rows u_i, v_i when i is even, and the copies of f when i is odd; add the two vertical edges obtained by holding the A-coordinate fixed and changing u_i to v_i. The selected edge in each block is then different at its two interfaces whenever both interfaces use the same physical row. Thus every cut is present, no edge of K_A is cut, and each operation joins two cycles into one. Induction on i gives a single cycle on all rows, including the one-block case. All added edges are actual one-coordinate changes in B. □

Lemma 5.21. *For a fixed star position, or no star, let two boundaries be equal or differ in one non-star position. Any overlapping component pair has at least ten common vertices.*

Proof. An unchanged constant component has 15 vertices. Between a constant and nonconstant boundary, one additional colour is excluded at one coordinate of that 15-cycle. A cyclic permutation of its three colours preserves each component: an order-three action on a two-element component set is trivial. Hence each colour appears five times at each coordinate and precisely ten states remain. In every other case, both components are their entire local state sets. The intersection lists have size at least three except possibly one list of size two. Start at that position, proceed around the cycle, and finally avoid the two already coloured neighbours. This gives at least $2 \cdot 2^3 \cdot 1 = 16$ colourings. With no changed position the bound only improves. Full lists cause no difficulty. □

Relative restoration maps. Call a coordinate and colour map Ψ a restoration map for a boundary t when $\Psi^{-1}(t)$ is standard; it restores the specified cycles, pools and protected sets of that standard type. Let t' equal t or differ from it in one non-star entry. Put $\Psi' = \Psi$ if $t' = t$, and otherwise $\Psi' = \Psi \circ N_u$, where $u = \Psi^{-1}(t')$ and N_u is the canonical restoration map of u. Transporting Claim 2 by Ψ shows that overlapping components at t and t', restored by Ψ and Ψ', have a common protected edge; the verifier also replays this rule over all labelled boundaries and admissible maps Ψ. The dependence of Ψ' on Ψ is necessary: for the labelled boundaries 00002 and 00003, both of type 00001, the two canonical maps restore protected sets with no common edge, although the components overlap.

Now let two consecutive layers carry maps related in this way in each factor; the two factors use independent maps and may change at the same base edge. If xx' is a common protected edge of overlapping first-factor components and $y_1, \dots, y_{10}$ are common states of overlapping second-factor components, then every $\{(x, y_i), (x', y_i)\}$ is an internal edge of both product cycles of Lemma 5.20

with the first factor as cycle factor, since these retain every protected edge in every row. Thus the two product cycles share at least ten internal edges.

Claim 3. *Let $h_0h_1\cdots h_{m-1}$ be a Hamilton path in the recolouring graph of the graph outside two pentagons. For $j \in \{1,2\}$, let $T_j(i)$ be the boundary list seen by the jth pentagon at h_i, and let $c_j(i) = 1$ when $F_{T_j(i)}$ is a constant two-component fibre, and $c_j(i) = 0$ otherwise. Form the graph whose vertices are the product components of $F_{T_1(i)}\square F_{T_2(i)}$, $0 \le i < m$, and whose edges record a common local state across h_ih_{i+1}, $0 \le i < m-1$. This component graph is connected.*

Proof. Give the two components of a constant fibre labels $0, 1$ and give a nonconstant fibre the single label $*$. A base edge changes at most one entry in either boundary list. If both consecutive fibres of a factor are constant, their boundary lists are equal and the label is preserved; otherwise, by Lemma 5.21, every label on one side meets every label on the other. The two factors follow these rules independently, and this describes the product component graph exactly.

For each j, some r_j has $c_j(r_j) = 0$. Otherwise every $T_j(i)$ is a constant word. Two constant words differing in at most one position are equal, so the whole Hamilton path would have one fixed boundary word. A global permutation of the four colours changing that word would give another proper colouring of the outside graph, which the Hamilton path must contain, a contradiction.

From any product component, walk layer by layer to r_1 and then to r_2, choosing label 0 whenever a constant fibre is entered from a nonconstant one. The first label is $*$ at r_1 and then evolves by this fixed rule, while the second label is $*$ at r_2. Hence every walk ends at the same component of layer r_2, and the graph is connected. □

Claim 4. *Let Q be a connected graph of maximum degree at most Δ. For each $v \in V(Q)$ let X_v be a vertex-disjoint fibre carrying a Hamilton cycle C_v. Suppose that, for every $uv \in E(Q)$, the two reference cycles share at least $r \ge \Delta$ internal edges, and each shared edge has the two corresponding cross edges in the ambient graph. Then the union of the fibres has a Hamilton cycle.*

Proof. Root a spanning tree of Q and process its vertices from the root outwards. Maintain one cycle on the fibres already processed; the only edges removed from a processed fibre are the internal edges used for its earlier tree attachments. When a child v is attached to its parent u, at most $\deg_Q(u) - 1 \le \Delta - 1$ shared edges at u have been removed. Since $r \ge \Delta$, a shared edge xy remains at u and is still present in the fresh cycle C_v. Delete xy from both cycles and add the two cross edges at x and y. This joins the two cycles into one and preserves the invariant. Each tree edge is processed once, so the final cycle covers every fibre. □

Restoration over a base cycle.

Lemma 5.22. *Suppose G has girth at least five and contains two vertex-disjoint 5-cycles with no edge between them. Each cycle has vertices of degree three except for at most one vertex of degree two. Let H be obtained by deleting both cycles. If $\mathcal{R}_4(H)$ is Hamiltonian, so is $\mathcal{R}_4(G)$. Neither planarity nor three-colourability of H is assumed here.*

Proof. Each degree-three cycle vertex has one exterior neighbour; a degree-two vertex gives a full list. An exterior vertex meets a given 5-cycle at most once, since two attachments would create a 3- or 4-cycle. Thus a bottom recolouring changes at most one list in each factor; the two factors may change simultaneously. All exterior neighbours remain in H, which is nonempty.

Delete one edge of a bottom Hamilton cycle to obtain a Hamilton path $h_0, \dots, h_{m-1}$. In each factor give $T_j(0)$ its canonical restoration map, and propagate maps along the path by the relative rule above. For every product component at h_i, take the product cycle of Lemma 5.20 restored by

the two maps at h_i, with the first pentagon as cycle factor. The component graph over the path is connected by Claim 3. It has at most four vertices per layer, hence maximum degree at most eight. Along each of its edges the two product cycles share at least ten internal edges. A cross edge at either endpoint of such an edge is the actual bottom recolouring, because the complete local colouring is held fixed while one exterior vertex changes colour. Claim 4 with $\Delta = 8$ and $r = 10$ gives one cycle through all extensions of all bottom colourings. □

5.9 Two separated five-faces

Lemma 5.23. *Let X be a connected plane graph that contains a cycle, has girth at least five, and has maximum degree at most three. If p and q are the numbers of vertices of degrees one and two, respectively, then*

$$f_5(X) = 12 - 4p - 2q + \sum_{\substack{F \text{ a face of } X \\ |F| \geq 7}} (|F| - 6). \tag{FC}$$

Here $f_5(X)$ counts facial walks of length five. Every facial walk has length at least five, and a facial walk of length five is a simple 5-cycle.

Proof. The boundary walk of a face is a closed walk. Since X is connected and contains a cycle, each such walk contains a simple cycle; a bridge, when it occurs on a boundary walk, is traversed twice on the way to and from the cyclic block. The girth assumption therefore gives length at least five. A non-simple closed walk contains a simple cycle and at least two further edge traversals, so in this graph it has length at least seven. Thus a length-five facial walk is a simple 5-cycle.

Put

$$L_X = \sum_{|F| \geq 7} (|F| - 6).$$

The face-length sum is $2e = 6f - f_5 + L_X$, while the degree sum is $2e = 3n - 2p - q$. Combining these equalities with Euler's identity $n - e + f = 2$ gives

$$f_5 = 6f + L_X - 2e = 12 - 6n + 4e + L_X = 12 - 4p - 2q + L_X,$$

as claimed. □

Lemma 5.24. *Every connected plane graph of girth at least five with $2 \leq d(v) \leq 3$ and at most one degree-two vertex contains two vertex-disjoint 5-cycles with no edge between them.*

Proof. We use Lemma 5.23 throughout. Set

$$q := |\{v \in V(G) : d_G(v) = 2\}| \in \{0, 1\}, \qquad p := |\{v \in V(G) : d_G(v) = 1\}| = 0.$$

If $q = 1$, let z denote the unique degree-two vertex; when $q = 0$ there is no vertex z in use. Thus G has at least $12 - 2q$ facial 5-cycles. If there is a bridge ab, take the side C not containing the unique degree-two vertex, if it exists, and relabel the endpoints so that $a \in C$. (If there is no degree-two vertex, choose either side.) In the graph induced by this side, deleting ab leaves a of degree two and every other vertex of degree three. It is cyclic: otherwise, being connected, it would be a tree, which is impossible when every vertex has degree at least two. Lemma 5.23 gives at least ten facial 5-cycles. Since a has two face angles, at most two of these cycles contain a; choose one, S, that avoids a.

Let D be the other side. If the original degree-two vertex is in $D \setminus \{b\}$, then D has two degree-two vertices; if it is b, then b is its only degree-one vertex; and if there is no such vertex, D has one

degree-two vertex. Thus $4p_D + 2q_D \le 4$ in all cases. The side D is cyclic, since a tree would have at least two degree-one vertices. The face-count lemma therefore gives at least eight facial 5-cycles in D. Any one of them is disjoint from S; it has no edge to S because ab is the only edge between the two sides and $a \notin S$.

Assume now there is no bridge. The subcubic graph is 2-connected: at a cutvertex every branch would require at least two incident edges to avoid a bridge, forcing degree at least four. Take a facial 5-cycle S and let B be its exterior neighbours. Girth makes them distinct; $b = |B|$ is five, or four when $q = 1$ and S contains the unique degree-two vertex z. Set $H = G - S$, embedded with the deleted region outside. Every member of B is on that outside face. A vertex outside S has at most one neighbour on S, by the girth assumption, and has degree at least two in G; hence H has no isolated vertex. Each member of B has degree two in H, except that, if $q = 1$ and $z \in B$, the vertex z becomes a leaf. There is at most one leaf, so every component of H is cyclic (a finite tree has at least two leaves).

In one such component, let r count its members of B, let q' count the degree-two vertices of G in the component that are not in B, and let ℓ be its outside facial length. Define

$$\varepsilon := \begin{cases} 1, & q = 1 \text{ and } z \in B \text{ lies in this component,} \\ 0, & \text{otherwise,} \end{cases} \qquad 0 \le q' \le 1.$$

When $q = 0$, these definitions give $\varepsilon = q' = 0$. The degree-one and degree-two counts in this component are

$$p_H = \varepsilon, \qquad q_H = r - \varepsilon + q'.$$

Applying Lemma 5.23 and removing the outside face gives

$$f_5^{\text{in}} = \ell + 6 - 2r - 2\varepsilon - 2q' + L, \qquad L = \sum_{\substack{F \text{ internal} \\ |F| \ge 7}} (|F| - 6) \ge 0. \tag{**}$$

Indeed,

$$4p_H + 2q_H = 4\varepsilon + 2(r - \varepsilon + q') = 2r + 2\varepsilon + 2q'.$$

The outside boundary is a closed walk containing a cycle, so $\ell \ge 5$. If $\ell = 5$ it contributes one facial 5-cycle, while if $\ell \ge 6$ it contributes $\ell - 6$ to the face-length excess; in either case the subtraction gives the displayed formula. A member of B belongs to at most one internal 5-face: its degree is at most two, with an outside face angle, and a leaf belongs to none. If H is disconnected, some component has $r \le 2$ since $|B| \le 5$. Since $\varepsilon + q' \le 1$ and $\ell \ge 5$, (**) gives

$$f_5^{\text{in}} \ge 5 + 6 - 2r - 2(\varepsilon + q') \ge 9 - 2r > r.$$

It therefore has more internal 5-faces than attachment vertices. One avoids B, so it is disjoint from S and has no edge to S.

Suppose H is connected. Let $F_1, \dots, F_t$ be the distinct faces on the exterior side of the edges from S to B. Deleting the five vertices of S and the $5 + b$ incident edges gives $n(H) = n(G) - 5$ and $e(H) = e(G) - 5 - b$; since H is connected, Euler's formula gives $f(H) = f(G) - b$. In the deletion, the 5-face bounded by S and the t exterior faces merge into the new outside face, so $t = b$. Thus the exterior faces are indeed distinct; call them $F_1, \dots, F_b$. Their merger has outside length

$$\ell = \sum_{i=1}^{b} |F_i| - (2b + 5), \tag{***}$$

because every edge of S occurs once among the F_i, while every edge from S to B occurs twice (once for each incident face).

If $q = 0$, any S has $b = 5$, $\ell \geq 10$, and $\varepsilon = q' = 0$. Formula (**) gives

$$f_5^{\text{in}} \geq 10 + 6 - 2 \cdot 5 = 6 > |B|.$$

If $q = 1$ and z lies on a 5-face, choose that face as S. Now $b = 4$, $\ell \geq 7$, and $\varepsilon = q' = 0$ (the unique degree-two vertex is on S, not in B). Hence

$$f_5^{\text{in}} \geq 7 + 6 - 2 \cdot 4 = 5 > |B|.$$

It remains that $q = 1$ and z lies on no 5-face. There is a 5-face adjacent to two distinct non-5-faces. To see this, consider the plane dual. It is loopless because G is bridgeless. Since G is 2-connected, its facial boundaries are cycles. Thus the faces of the dual are triangles, one for each degree-three vertex of G, except for the unique digon corresponding to z. The digon's endpoints are distinct (there is no cutvertex) and represent the two faces incident with z; both are non-5-faces because z lies on no 5-face. Let W be the set of dual vertices representing non-5-faces. Suppose, for a contradiction, that every dual vertex outside W has at most one distinct neighbour in W. Then in every triangular dual face whose two consecutive vertices lie in W, the third vertex also lies in W; otherwise it would have two distinct neighbours in W.

Start with the digon edge. Its endpoints lie in W. Walk around either endpoint in the fixed plane embedding. Each intervening face is triangular, except possibly the unique digon, so the preceding paragraph shows inductively that every successive neighbour is in W; the digon only repeats the same neighbour. The same argument applies at every newly reached vertex of W. Hence W is closed under adjacency. The dual is connected, so this would give $V(G^*) = W$, contradicting the at least ten 5-faces of G, whose dual vertices are not in W. Therefore some 5-face is adjacent to two distinct non-5-faces, as asserted. Choose the resulting 5-face S. If its deletion disconnects the graph, use the case already proved; otherwise $b = 5$ and two distinct neighbours in (***) have length at least six. Thus $\ell \geq 12$. If $q = 1$ and $z \in B$, then $\varepsilon = 1, q' = 0$; if $q = 1$ and $z \notin B$, then $\varepsilon = 0, q' \leq 1$. In both cases $\varepsilon + q' \leq 1$, so (**) gives

$$f_5^{\text{in}} \geq 12 + 6 - 2 \cdot 5 - 2 = 6 > |B|.$$

In every case one avoids B, making it disjoint and nonadjacent to S. □

Completing the subcubic induction.

Proof of Theorem 5.19. Induct on the number of vertices, allowing disconnected graphs. The one-vertex case is $\mathcal{R}_4(G) = K_4$. A vertex of degree at most one is handled by Lemma 2.2. If there are two degree-two vertices, adjacent ones use Lemma 5.5. Ones at distance two use Lemma 5.10, unless their middle vertex has degree two, when an adjacent pair already applies. Ones at distance at least three use Lemma 5.6, whose spare-colour witness uses Grötzsch's theorem [4]; the new pentagon lift does not. Pairs in different components have disjoint neighbourhoods and are covered by the same separated-pair argument.

The remaining graph has minimum degree at least two and at most one degree-two vertex. Apply Lemma 5.24 inside any component. Delete its two separated pentagons. The remaining graph is nonempty and belongs to the same class, so induction gives its four-colour Hamilton cycle. Lemma 5.22 restores both pentagons. This proves the assertion at four colours. For every $k \geq 5$ use the triangle-free planar theorem proved earlier, with larger colours separately supplied there by degeneracy. The frozen 012012 colouring of C_6 gives the Hamiltonian threshold lower bound four. □

5.10 Graphs with subcubic blocks

Theorem 5.25. *Let G be a nonempty finite simple planar graph of girth at least five. If every block B satisfies $\Delta(B) \leq 3$, then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. Cutvertices may have arbitrarily large degree in G. The uniform Hamiltonian threshold of this class is exactly four. Its Hamilton-connected threshold lies between four and five.*

Proof. Treat bridges as two-vertex blocks and isolated vertices as singleton blocks. The class is closed under vertex deletion: a nontrivial two-connected block of a subgraph is a two-connected subgraph of G, hence is contained in one original block, with no larger internal degrees. Planarity and girth are also preserved.

Products of Hamiltonian graphs are Hamiltonian: copy one factor cycle along a spanning path of another factor cycle, and merge successive copies using two alternating, distinct internal edges. This also handles disconnected original graphs. Choose a smallest four-colour counterexample, which may therefore be assumed connected. The singleton base is K_4, and the single-vertex lift excludes vertices of degree at most one. If there is only one block, the previous subcubic girth-five theorem applies. Thus assume minimum degree at least two and at least one cutvertex.

Lemmas 5.5, 5.10, and 5.6 exclude adjacent degree-two vertices, a $(2, 3, 2)$ path, and two degree-two vertices at distance at least three. Consequently any two degree-two vertices not already giving a reduction must have a common neighbour of degree at least four in G.

The block-cut tree has at least two endblocks. No endblock is a bridge, since its noncut endpoint would be a leaf. Let B be an endblock and x its unique cutvertex of G. Every vertex of $B - x$ has its full G-degree in B, hence degree two or three; also $d_B(x) \in \{2, 3\}$. Let q count the degree-two vertices of $B - x$ and put $\varepsilon = [d_B(x) = 2]$. If $q \geq 2$, any pair of these vertices has a common neighbour of G-degree at least four by the excluded reductions. Their neighbours all lie in B, so only x can be this common neighbour. Every one of the q vertices is therefore adjacent to x, giving $q \leq d_B(x) = 3 - \varepsilon$. Together with the immediate $q \leq 1$ case, this proves $q + \varepsilon \leq 3$.

Euler's identity in the two-connected block B now gives

$$f_5(B) = 12 - 2(q + \varepsilon) + \sum_{|F| \geq 7} (|F| - 6) \geq 6.$$

Each five-face is a simple pentagon, while at most $d_B(x) \leq 3$ faces meet x, so choose one avoiding x. If $q \leq 1$, it contains at most one degree-two vertex of G. If $q \geq 2$, a cycle through any of the q vertices must use its edge to x, so the chosen pentagon contains none. All its other vertices have G-degree three.

Choose these pentagons in two distinct endblocks. Their vertices are noncutvertices of G and belong to their respective endblocks only, so the pentagons are disjoint and have no edge between them. The established protected-edge lift for two separated pentagons applies, including its allowance of one degree-two vertex per pentagon and simultaneous exterior list changes. The smaller graph remains in the class and has a bottom Hamilton cycle by induction. Restoration contradicts minimality.

For every $k \geq 5$ separately use the established triangle-free planar theorem. Finally C_6 belongs to the class and its periodic 012012 colouring is frozen in $\mathcal{R}_3(C_6)$, giving the lower bound four. □

In particular, a remaining minimum counterexample for general planar girth-five four-colour Hamiltonicity must contain a block with a vertex of degree at least four *inside that block*. High total degree accumulated only at cutvertices is insufficient. This does not assert that every minimum counterexample is two-connected.

5.11 One exceptional vertex

Lemma 5.26. *Let G be a connected plane graph of girth at least five. Suppose one vertex w has degree $d \geq 4$ and every other vertex has degree three. Then G contains two vertex-disjoint 5-cycles with no edge between them, all of whose vertices have degree three in G.*

Proof. For a connected plane graph X of girth at least five and minimum degree at least two, write $s_X = |\{v : d_X(v) = 2\}|$, $D_X = \sum_{d_X(v)\geq 4}(d_X(v) - 3)$, and $Q_X = \sum_{|F|\geq 7}(|F| - 6)$. Facial lengths count edge sides. The degree and face sums give

$$f_5(X) = 12 + Q_X + 2D_X - 2s_X,$$

where every facial walk of length five is a simple cycle.

Suppose ab is a bridge, with a on the side A avoiding w. Only a has degree two in A, so A has at least ten pentagonal faces, at most two containing a. Choose one, S, avoiding a. On the other side, if $b \neq w$, only b becomes degree two and w retains degree d. There are at least $4 + 2d$ pentagonal faces, at most d containing w. If $b = w$, its remaining degree is $r = d - 1 \geq 3$, and there are at least $6 + 2r$ pentagonal faces, at most r containing w. In either case choose a pentagon T avoiding w. All vertices of S and T have original degree three. They lie on opposite sides of the bridge, and S avoids its endpoint, so no edge joins them.

Assume there is no bridge. At a cutvertex, each component left by its deletion uses at least two incident edges; otherwise its sole attachment edge would be a bridge. Thus the only possible cutvertex is w. Restore w and its incident edges into each component of $G - w$. Its degree r in that branch is at least two. The branch has at least ten pentagonal faces if $r = 2$, and at least $6 + 2r$ if $r \geq 3$. In both cases more than r pentagons exist, so choose one avoiding w. Two branches give the required pair.

We may therefore assume G is two-connected. Its facial boundaries are cycles, and the faces at each vertex are distinct. In the plane dual, every face is a triangle with three distinct vertices except the single d-gon corresponding to w, whose vertices are also distinct. Parallel dual edges are allowed. Let W consist of the dual vertices representing nonpentagonal primal faces or faces containing w. There is a vertex outside W: the face identity gives at least $6 + 2d$ pentagonal faces, at most d containing w.

Some vertex outside W has two distinct neighbours in W. Otherwise every triangular dual face with two vertices in W has its third in W. All boundary vertices of the exceptional polygon belong to W. At such a vertex, removing its one exceptional corner leaves a path through all neighbour positions in the rotation. Starting with a boundary neighbour in W, the triangular-face rule propagates in both directions along that path, putting every neighbour in W. Off the exceptional polygon, all corners are triangular, so the same rule propagates around the entire rotation from any known neighbour in W. Connectedness of the dual would put every vertex in W, a contradiction. The argument uses neighbour positions and remains valid with parallel edges.

Choose the resulting ordinary facial pentagon S, which has two distinct neighbouring faces in W. Its five exterior neighbours form a set B of size five, by girth. Put $H = G - V(S)$; it has minimum degree at least two. If H is disconnected, every component has at least two attachment edges to S, since G has no bridge. There are five such edges, so there are exactly two components. The component C avoiding w has $r \leq 3$ attachments. All its attachment vertices have degree two; all other vertices have degree three. The connected deleted region lies in one face of C, which we choose as the outside face. Every attachment has an outside angle. If its boundary length is $\ell \geq 5$ and $L \geq 0$ is the excess over six of its internal long faces, the degree and face sums give

$$f_5^{\text{in}}(C) = \ell + 6 - 2r + L \geq 11 - 2r > r.$$

Each attachment has at most one internal angle, hence meets at most one internal pentagon. Some internal pentagon avoids B and gives the desired T.

Suppose H is connected. The five exterior neighbour faces F_i of S are distinct. Indeed, deleting five vertices and ten edges reduces the face count by five. The pentagonal face and all its exterior neighbours merge into one face, so there must have been six distinct incident faces. Put

$$\varepsilon = \sum_{i=1}^{5} (|F_i| - 5) \geq 0.$$

Choose the merged face as the outside face of H. Its length is

$$\ell = \sum_{i=1}^{5} |F_i| - 15 = 10 + \varepsilon :$$

the sum loses five cycle-edge sides and both sides of each of the five attachment edges.

Let $\beta = [w \in B]$, let a count outside angles at w in H, and let L be the internal excess over six. Here $s_H = 5 - \beta$ and $D_H = d - \beta - 3$. Since $\ell \geq 10$, the outside face contributes $\ell - 6$ to Q_H and is not pentagonal. The face identity therefore gives

$$f_5^{\mathrm{in}}(H) = 12 + \ell - 6 + L + 2(d - \beta - 3) - 2(5 - \beta) = 2d + \varepsilon + L.$$

Each vertex of $B \setminus \{w\}$ has degree two and an outside angle, so at most $5 - \beta$ internal pentagons meet those vertices. At most $d - \beta - a$ internal pentagons meet w. The forbidden total is thus at most $d + 5 - 2\beta - a$.

If $\beta = 1$, then $a \geq 1$ and the internal pentagon count exceeds this bound. If $\beta = 0$, use the two distinct neighbouring faces of S in W. Each nonpentagonal one contributes at least one to ε. Each remaining one contains w and contributes an outside angle at w: no edge at w was deleted, so its corner survives intact. Distinct faces give distinct angles. Consequently $\varepsilon + a \geq 2$, and

$$(2d + \varepsilon + L) - (d + 5 - a) \geq d - 3 \geq 1.$$

An internal pentagon T therefore avoids B and w. Its vertices have original degree three, and no edge joins it to S. □

Theorem 5.27. *For every nonempty planar graph G of girth at least five with at most one vertex of degree at least four, $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. The uniform Hamiltonian threshold of this class is four.*

Proof. The class is closed under vertex deletion. Take a vertex-minimal counterexample at four colours. By Lemma 5.16, it is connected and has minimum degree at least two. Lemma 5.17 excludes degree-two vertices, since their presence would require at least four vertices of degree at least four. If no vertex is heavy, apply Theorem 5.19. Otherwise every vertex has degree three except one of degree at least four.

Lemma 5.26 supplies two vertex-disjoint, nonadjacent ordinary pentagons. Delete them and apply induction; the exceptional vertex remains, so the deleted graph is nonempty. Lemma 5.22 restores the two cycles, contradicting minimality.

For $k \geq 5$, use Theorem 4.5. The periodic three-colouring 012012 of C_6 is frozen, and C_6 belongs to the class, proving the lower bound four. □

5.12 Two-degenerate subclasses

Proposition 5.28. *A minimum counterexample in the sense of Lemma 5.16 with at most seventeen distinct 5-cycles has no degree-two vertex.*

Proof. The case $s = 1$ requires $p \geq 20$. If $s \geq 2$, then $p \leq 17$ and the budgets with $t \geq 1$ allow exactly

$$(t, D', h', Q, R) \in \{(1,3,3,0,0), (2,3,3,0,0), (1,4,4,0,2), (1,4,4,1,1)\}.$$

Lemma 5.18 makes every face boundary simple and every face length at most seven. Call an angle at w between two edges to S an S–S angle.

In the first two cases, $Q = R = 0$ and the three other heavy vertices have degree four. The path argument in the proof of Theorem 5.4 excludes every S–S angle: its pentagonal face gives a regular or one-heavy order-four path, and its hexagonal face gives a regular order-five path. Thus $s \leq t$. Since $s \geq 2$ and $t \leq 2$, the only remaining possibility is $s = t = 2$ and $d(w) = 4$, with the neighbours alternating between S and Z. The equality $p_0 = 2t = 4$ in (5.5) gives four pentagonal faces containing w and no other heavy vertex. They are all four faces incident with w.

Choose one using u_1. The other degree-two neighbour u_2 is outside that simple cycle: otherwise its edge to w would also be a cycle edge, contradicting the chosen angle. Its other neighbour x_2 is also outside, since every vertex of a pentagon is at cycle distance at most two from w, and the path w, u_2, x_2 would create a triangle or 4-cycle. The face together with u_2 is the tailed pentagon of Lemma 5.13, a contradiction.

In the last two cases, $t = 1$, $R + Q = 2$, and all four other heavy vertices have degree four. If $s = 3$, the first case of Lemma 5.13 applies. Otherwise $s \geq 4$, so there are $s - 1 \geq 3$ distinct S–S faces. A pentagonal one with at most one other heavy vertex gives a regular or one-heavy order-four path. Otherwise it contributes at least one to the last sum of (5.5). A hexagonal one with at most one other heavy vertex gives a regular or one-heavy order-five path, covered by Lemmas 5.12 and 5.13. Otherwise it contributes at least two to the nonpentagonal-incidence sum in (5.5). A heptagonal face contributes one to Q. Contributions from distinct faces are disjoint, so the three faces require at least three units from $R + Q = 2$, a contradiction. □

Corollary 5.29. *Let G be a nonempty two-degenerate planar graph of girth at least five. If G has at most three heavy vertices or at most seventeen distinct 5-cycles, then $\mathcal{R}_k(G)$ is Hamiltonian for every $k \geq 4$. Each of these two classes has uniform Hamiltonian threshold four.*

Proof. Both classes are closed under vertex deletion: degrees cannot increase, and two-degeneracy and the cycle-count bound are preserved. A minimum four-colour counterexample has minimum degree at least two by Lemma 5.16, and thus has a degree-two vertex by two-degeneracy. Lemma 5.17 excludes the class with at most three heavy vertices, and Proposition 5.28 excludes the other class. For $k \geq 5$, Theorem 2.5 applies. The graph C_6 belongs to both classes and its frozen three-colouring gives sharpness. □

6 The girth-five frontier

The threshold intervals in Table 1.1 leave two questions: does every $G \in \mathcal{P}_5$ have a Hamiltonian $\mathcal{R}_4(G)$, and is every such recolouring graph Hamilton-connected? The latter question also remains open for the four-colour subclasses of Theorem 1.3. Known connectivity [1] does not settle either question. Theorems 5.4, 5.27 and 5.25 show that a minimum Hamiltonian counterexample must have at least sixteen distinct 5-cycles, at least two vertices of degree at least four, and a block with internal maximum degree at least four.

Remark 6.1. Suppose a graph of minimum order with non-Hamiltonian $\mathcal{R}_4(G)$ in the full class $\mathcal{P}_5$ has at most seventeen distinct 5-cycles. Proposition 5.28 gives minimum degree at least three. Equation (5.1) gives $D \leq 2$, and Theorem 5.27 excludes at most one heavy vertex. Thus G has exactly two degree-four vertices and all other vertices have degree three, with $p = 16 + Q$ and $Q \leq 1$. By Lemma 5.18, all faces have simple boundaries of length five or six except possibly one face of length seven. The graph is two-connected, since a cutvertex of a connected plane graph occurs repeatedly on a facial walk. This necessary condition leaves the two-degree-four-vertex case unresolved and does not assert that a counterexample exists.

7 Finite certificates and reproducibility

The supplementary source `article_certificates.tex` supplies the adjacent-pair, $(2, 3, 2)$ path and pentagon data. The literal cycles for paths with a common outside neighbour are supplied in `path4-family.json` and `path5-family.json` under `supplement/data/`. The seven further degree-two families are in `supplement/data/degree-two/`. Three pentagonal families are in `supplement/data/pentagonal-reservoirs/`. Both sets have separate direct checks of their selected-cycle interfaces in physical colour labels. The eight exact readers reconstruct the local graphs and their stated interfaces. Missing blocks, duplicate records, illegal transitions, incomplete state coverage and failed protection conditions cause the check to fail. No solver output or earlier computation report is an input to these checks.

Table 7.1: Finite checks supporting the local restoration lemmas.

Supplement	Checked objects and coverage
Appendix A	Seven edge-covering cycle orbits; 53 endpoint words with exact subset-DP cross-check; 225 labelled list pairs and 3,225 interfaces.
Appendix B	Three normalized path cycles; 60 good boundaries and all 504 directed adjacent good-boundary changes; safe edges for every allowed cycle pair and support at bad neighbours.
Appendices C–D	23 complete pentagon components; 2,304 labelled boundaries; ordered path blocks; both hard two- and three-row seeds; pool membership and protection in every row; all 313 interfaces, partitioned as $47 + 236 + 30$; all 23^2 product cycles with the alternating reserve splice replay; and the relative-normalization rule on all labelled boundaries and admissible maps (75,120 overlapping cases).
Appendix E	Five order-four and fifteen order-five path cycles; all 64 and 256 labelled boundaries and all 576 and 3,072 directed physical-boundary interfaces; at least three and five common chosen-cycle edges, respectively.
Appendix F	Seven degree-two cycle families with 421 templates and 96,036 template states; all 8,704 labelled boundaries and 158,976 directed interfaces, including every allowed paired-port move; a separate physical replay of all interface inequalities.
Appendix G	Three pentagonal families with 781 templates, 786 component cycles and 101,121 template states; all 17,664 labelled boundaries and 486,912 directed interfaces. Every overlapping component pair shares at least four selected-cycle edges in the reservoir and at least two in the other models; every component has an overlapping partner on both sides. A separate physical replay checks all interface inequalities.

Run `python3 supplement/verify_all.py` from the article directory. For the standalone delivery directory, add `-check-manifest` to verify every listed file byte for byte. The scripts require only the Python standard library and use exact integer and finite-set operations. The runner also checks rejection of corrupted cycles, missing certificate blocks, invalid interface edges and altered selector

data. Its JSON reports record input hashes and coverage; the distribution includes commands for rebuilding both PDFs and a file manifest.

The finite verification establishes the local certificate claims, the finite 23^2 product/reserve replay and the relative-normalization replay in Table 7.1. The path lifting, relative normalization and component-graph connectivity along an arbitrary bottom Hamilton path, spanning-tree splicing and structural inductions are analytic arguments in the article. Thus the finite checks have no bound on the order of G as an input premise, and the general girth-five questions stated above remain open.

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